MATH3404

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Chapter 1

Optimization in $\mathbb{R}^n$

1.1 Functions of One Variable

Problem I. Let $f(x)$ be a function on some interval I of the real line $\mathbb{R}$. Find the points of I at which $f(x)$ achieves its maximum and minimum values.

Definition 1.1.1

If

$$f(\bar{x}) \leq f(x); \quad \text{for all } x \in I, \tag{1.1}$$

with equality for $x = \bar{x} \in I$, then $f(x)$ achieves its minimum value at $\bar{x}$.
(1.1) means that $f(x)$ achieves its global (or absolute) minimum at the point $\bar{x}$.

Using $g(x) = -f(x)$, we have a similar definition for a maximum, i.e. $f(\bar{x}) \geq f(x)$ for all $x \in I$.

Definition 1.1.2

$f(x)$ has a local minimum at a point $\bar{x}$ if

$$f(\bar{x}) \leq f(x); \quad \text{for all } x \in N \subset I, \tag{1.2}$$

with equality for $x = \bar{x}$, where N is the ε-neighborhood $|x - \bar{x}| < \varepsilon$, with ε small.

Theorem 1.1.3

Let $f(x)$ be defined on an open interval $I = (a, b)$, $a < b$ and let f be C^2 (continuous second derivative) in some ε-neighborhood of $\bar{x} \in (a, b)$. If $f(x)$ has a local minimum at $\bar{x}$, then $f'(\bar{x}) = 0$.

Proof. If $f(\bar{x})$ is a local minimum, then there exists a neighborhood N such that

$$f(\bar{x} + h) - f(\bar{x}) \geq 0, \quad \forall\, \bar{x} + h \in N,$$

with equality for $h = 0$. Using Taylor's expansion, we have

$$f(\bar{x} + h) = f(\bar{x}) + hf'(\bar{x}) + \frac{h^2}{2!} f''(\bar{x} + \theta h), \qquad 0 < \theta < 1,$$

where $2! = 1 \times 2 = 2$. Then we have

$$hf'(\bar{x}) + \frac{h^2}{2!} f''(\bar{x} + \theta h) \geq 0 \quad \text{in } N,$$

where $f''(\bar{x}+\theta h)$ is bounded. When $h>0$ we can deduce that

$$f'(\bar{x})+\frac{h}{2}f''(\bar{x}+\theta h)\geq 0.$$

As $h\to 0$, we obtain $f'(\bar{x})\geq 0$. When $h<0$, a similar argument yields $f'(\bar{x})\leq 0$, so $f'(\bar{x})=0$ is a necessary condition for a local minimum. □

We have two theorems on sufficient conditions of a local minimum.

Theorem 1.1.4

Let $f(x)$ be defined on an open interval $I=(a,b)$ and let f be C^1 (continuous first derivative) in some ε-neighborhood of $\bar{x}\in(a,b)$. If $f'(\bar{x})=0$, then $f(x)$ has a local minimum at $\bar{x}$ provided $f'(\bar{x}+h)<0$ for $h<0$ and $f'(\bar{x}+h)>0$ for $h>0$.

Proof. Omitted. Hint: using the mean–value theorem, we have

$$f(\bar{x}+h)=f(\bar{x})+hf'(\bar{x}+\theta h),\quad 0<\theta<1,$$

where $2!=1\times 2=2$. □

Example.

Let $f(x)=x^2$ be a function on $(-1,1)$. $f'(x)=2x$ so $\bar{x}=0$ is a point satisfying all conditions in Theorem 1.1.4. Therefore f has a local minimum at $\bar{x}=0$.

Theorem 1.1.5

Let $f(x)$ be C^m (continuous first m-derivative with integer $m\geq 1$) in some ε-neighborhood of $\bar{x}\in(a,b)$. If

$$f'(\bar{x})=f''(\bar{x})=\cdots=f^{(m-1)}(\bar{x})=0,\quad f^{(m)}(\bar{x})\neq 0,$$

then $f(x)$ has a local minimum at $\bar{x}$ provided (i) m is even and (ii) $f^{(m)}(\bar{x})>0$.

Proof. The Taylor expansion yields

$$f(\bar{x}+h)-f(\bar{x})=hf'(\bar{x})+\cdots+\frac{h^{m-1}}{(m-1)!}f^{(m-1)}(\bar{x})+\frac{h^m}{m!}f^{(m)}(\bar{x}+\theta h),\qquad 0<\theta<1,$$

where $m!=1\times 2\times\cdots\times m$. By assumption in Theorem 1.1.4, we have

$$f(\bar{x}+h)-f(\bar{x})=\frac{h^m}{m!}f^{(m)}(\bar{x}+\theta h).$$

Since $f^{(m)}$ is continuous near $\bar{x}$ and $f^{(m)}(\bar{x})>0$, we know that $f^{(m)}(\bar{x}+\theta h)>0$ for sufficiently small h. Thus $f(\bar{x}+h)-f(\bar{x})>0$ for small h. This means that f has a local minimum at $\bar{x}$.

□

1.2 Critical Points, End-points and Points of Discontinuity

In the last section, we need to understand the point $\bar{x}$ with $f'(\bar{x})=0$. But sometime $f'(\bar{x})$ does not exists.

Example.

$f(x)=|x|$, with $-\infty<x<\infty$, has a global minimum at $x=0$, but $f'(0)$ doesn't exist.

We need to extend the theory to the case that $f(x)$ is not differentiable.

Definition 1.2.1

We use the term *critical point* to mean a point at which either $f'(x) = 0$ or $f'(x)$ does not exists.

Theorem 1.2.2

Let $f(x)$ be defined on (a, b) and differentiable on (a, b) except at $\bar{x}$. Consider a deleted neighborhood of $\bar{x}$, that is the set of points $\bar{x} - \varepsilon < x < \bar{x}$ and $\bar{x} < x < \bar{x} + \varepsilon$ for some small positive ε. Then $f(x)$ has a local minimum at $\bar{x}$ provided $f'(x) < 0$ through $\bar{x} - \varepsilon < x < \bar{x}$ and $f'(x) > 0$ through $\bar{x} < x < \bar{x} + \varepsilon$.

Proof. Using the mean–value theorem, we have

$$f(\bar{x} + h) = f(\bar{x}) + hf'(\bar{x} + \theta h), \qquad 0 < \theta < 1.$$

□

The natural of the interval I is important too. When $f(x)$ is defined on a closed interval $[a, b]$, then $f(x)$ has a global maximum or minimum on I, possible achieve at the end points a and b.

Example.

> Let $f(x) = x$ on $[0, 2]$ has a minimum at 0 and maximum at 2.

Example.

> Consider $f(x) = x^3$ on $I = (-1, 2)$; it has neither a global maximum nor a minimum.

If $I = [-1, 2]$, then the function has a global minimum at $x = -1$ and a global maximum at $x = 2$.

We shall assume that any $f(x)$ we deal with is piecewise continuous. If this is the case, isolated points of discontinuity also need to be examined, because they too could be global maxima or minima.

Example.

> Consider
>
> $$f(x) = \begin{cases} -x^3, & -\infty < x < 1, \\ x - 3, & 1 \leq x < \infty. \end{cases}$$

This function has a global minimum at $x = 1$, where $f(x) = -2$.

Assume that f and f' are continuous on I except at a finite number of points where $f'(x)$ does not exist or $f(x)$ is discontinuous. The procedure to find a global minimum is

1. Find the stationary points ($f'(x) = 0$) and use Theorem 1.1.4 to determine any local minima.
2. Examine $f(x)$ near every point at which $f(x)$ or $f'(x)$ is discontinuous to see if it is a local minimum.
3. Compare all these values of $f(x)$ to find the smallest.
4. If I is closed, evaluate $f(a)$ and $f(b)$ and compare them with the smallest value found in (iii). This gives the global minimum, if it exists.

Example.

> Find the global minimum (when it exists) of the following function: $f(x) = 2x^3 - 9x^2 + 12, \quad -\infty < x < \infty$

1.3 Functions of Several Variables

In this section, we turn to deal with minimizing problem of a function of several variables.

Let $x = (x_1, x_2, \ldots, x_n)$ be n variables.

Problem II. Let $f(x)$ be a function on some region K of $\mathbb{R}^n$. Find the points of K at which $f(x)$ achieves its maximum and minimum values.

Definition 1.3.1

Let $f(x)$ be a given function of $x = (x_1, \ldots, x_n)$ defined in K. We say that f has a global minimum at $x = a$, with $a \in K$ if

$$f(a) \leq f(x); \quad \text{for all } x \in K, \tag{2.1}$$

with equality only for $x = a$.

Similarly, we define a local minima. Let N be an ε_0-neighborhood of $a = (a_1, \ldots, a_n)$ (i.e. $y = (y_1, \ldots, y_n) \in N$, $|y - a| = \sqrt{|y_1 - a_1|^2 + \cdots + |y_n - a_n|^2} \leq \varepsilon_0$).

For any point y in N and any nonzero vector h, there is some sufficiently small ε such that $y = a + \varepsilon h$.

Definition 1.3.2

Let $f(x)$ be a given function of $x = (x_1, \ldots, x_n)$ defined in K. We say that f has a local minimum at $x = a$, with $a \in K$ if

$$f(a) \leq f(y) \quad \text{for all } h \text{ and } \varepsilon \text{ with } y = a + \varepsilon h \in N, \tag{2.2}$$

Now assume that in N, f has continuous first and second order partial derivatives with respect to all its variables and that the third order derivatives are bounded.

Definition 1.3.3

For $a \in \mathbb{R}^n$, the gradient of $f(x)$ at a is

$$\operatorname{grad} f(a) = \left(\tfrac{\partial f}{\partial x_1}(a), \ldots, \tfrac{\partial f}{\partial x_n}(a)\right)^T.$$

Definition 1.3.4

The Hessian matrix $H(a)$ of $f(x)$ is the $n \times n$ matrix whose entries are the second derivatives of f evaluated at a; i.e.

$$H(a) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)_{n \times n}.$$

Let h be a vector in $\mathbb{R}^n$ (as an $n \times 1$-matrix), i.e.

$$h = \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix}.$$

Then $h^T = (h_1, \ldots, h_n)$ is the transpose of h. Note that

$$h^T \operatorname{grad} f(a) = \sum_{i=1}^{n} h_i \frac{\partial f}{\partial x_i}(a) \quad \text{and} \quad h^T H(a) h = \sum_{i,j=1}^{n} h_i \frac{\partial^2 f}{\partial x_i \partial x_j} h_j.$$

Using Taylor series, we have

$$f(a + \varepsilon h) = f(a) + \varepsilon h^T \operatorname{grad} f(a) + \frac{\varepsilon^2}{2} h^T H(a) h + O(\varepsilon^3),$$

for all $h \in \mathbb{R}^n$ and sufficiently small ε.

Theorem 1.3.5

If f has a local minimum at $x = a$ inside a domain K and continuous second order derivatives, and the third order derivative of f is bounded, then

$$\nabla f(a) := \operatorname{grad} f(a) = 0.$$

Proof. Since a is a local minima,

$$f(a + \varepsilon h) \geq f(a)$$

for all $h \in \mathbb{R}^n$ and sufficiently small ε such that $a + \varepsilon h \in N$. Assume $|h| = 1$ and $|\varepsilon| \leq \varepsilon_0$ for a small $\varepsilon_0 > 0$.

By the above Taylor expansion, we have

$$f(a + \varepsilon h) - f(a) = \varepsilon h^T \operatorname{grad} f(a) + \frac{\varepsilon^2}{2} h^T H(a) h + O(\varepsilon^3) \geq 0.$$

Dividing by $\varepsilon > 0$ and letting $\varepsilon \to 0^+$,

$$h^T \operatorname{grad} f(a) \geq 0.$$

Similarly, dividing by $\varepsilon < 0$ and letting $\varepsilon \to 0^-$,

$$h^T \operatorname{grad} f(a) \leq 0.$$

Therefore

$$h^T \operatorname{grad} f(a) = 0$$

for $h \in \mathbb{R}^n$ with $|h| = 1$. Then $\operatorname{grad} f(a) = 0$. Otherwise, choose $h = \dfrac{\operatorname{grad} f(a)}{|\operatorname{grad} f(a)|}$. □

Example.

Let $f(x_1, x_2) = x_1^2 + x_2^2$ be defined on $\mathbb{R}^2$ and have a local minimum at $(0, 0)$. Then $\operatorname{grad} f(0, 0) = 0$.

Theorem 1.3.6

Let f be defined on an open domain $\Omega \subset \mathbb{R}^n$ and have continuous second-order derivatives. A sufficient condition for $f(x)$ to have a local minimum at a point $a \in \Omega$ is that

$$\operatorname{grad} f(a) = 0 \quad \text{and} \quad h^T H(a) h > 0, \qquad \forall h \in \mathbb{R}^n.$$

Proof. By Taylor's expansion,

$$f(a + \varepsilon h) = f(a) + \varepsilon h^T \nabla f(a) + \frac{\varepsilon^2}{2} h^T H(a) h + O(\varepsilon^3 |h|^3).$$

For $x = a + \varepsilon h \in B_\varepsilon(a)$ and for all $h \in \mathbb{R}^n$ with $|h| = 1$, we obtain

$$f(x) - f(a) = f(a + \varepsilon h) - f(a) = \frac{\varepsilon^2}{2} h^T H(a) h + O(\varepsilon^3 |h|^3) \geq \min\{\lambda_i\} \frac{\varepsilon^2}{2} |h|^2 + O(\varepsilon^3 |h|^3) > 0$$

for $|\varepsilon| \leq \varepsilon_0$ with a sufficiently small ε_0, where λ_i is an eigenvalue of $H(a)$. □

Theorem 1.3.7

The quadratic form $h^T H h > 0$ is positive if and only if $\det H$ and all principal minors of H are positive (this means that all eigenvalues of H are positive).

Proof. Omitted. □

In $\mathbb{R}^3$, this would require

$$\frac{\partial^2 f}{\partial x_1^2} > 0, \quad \det \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{pmatrix} > 0, \quad \text{and} \quad \det \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_3} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \frac{\partial^2 f}{\partial x_2 \partial x_3} \\ \frac{\partial^2 f}{\partial x_3 \partial x_1} & \frac{\partial^2 f}{\partial x_3 \partial x_2} & \frac{\partial^2 f}{\partial x_3^2} \end{pmatrix} > 0.$$

The corresponding theorem for a local maximum requires that $\nabla f = 0$ and $h^T H h < 0$ for all h.

Theorem 1.3.8

A quadratic form $h^T H h$ is negative if and only if $(-1)^n \det H > 0$ and the principal minors of H alternate in sign with $\frac{\partial^2 f}{\partial x_1^2} < 0$ (this means that all eigenvalues of H are negative).

Theorem 1.3.9

Let f be defined on an open domain $\Omega \subset \mathbb{R}^n$ and have continuous second-order derivatives. A sufficient condition for $f(x)$ to have a local maximum at a point $a \in \Omega$ is that

$$\nabla f(a) = 0 \quad \text{and} \quad h^T H(a) h < 0, \qquad \forall h \in \mathbb{R}^n.$$

Proof. Take $g(x) = -f(x)$. Then we can apply Theorem 1.3.6. □

Example.

> Minimize $f(x_1, x_2) = x_1^2 - x_2^2$ where Ω is $\mathbb{R}^2$.

$\operatorname{grad} f = (2x_1, -2x_2)^T$ and this is zero for $x_1 = x_2 = 0$ and

$$\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$$

Now $\det H = -4$ and $\dfrac{\partial^2 f}{\partial x_1^2} = 2$, so the quadratic form is neither positive nor negative and the origin is neither a maximum nor a minimum.

Example.

> Find the local maxima and minima of the functions in $\mathbb{R}^2$:
>
> 1. $f(x_1, x_2) = (x_1^2 - 4)^2 + x_2^2$.
> 2. $f(x_1, x_2) = x_1^3 + x_3^3 + 3x_1^2 - 3x_2^2 - 8$.

1.4 Minimization with Constraints

Problem III. Minimize $f(x)$ in $\mathbb{R}^n$ given that x satisfies the equality constraints

$$g_j(x) = c_j, \qquad j = 1, \ldots, m < n,$$

where $c_1, \ldots, c_m$ are given numbers.

Theorem 1.4.1

Let $f(x)$ and $g_j(x)$ be defined and have continuous second derivatives in some open region of $\mathbb{R}^n$. Then a necessary condition that a minimize $f(x)$ with the constraints

$$g_j(x) = c_j, \qquad j = 1, \ldots, m < n$$

is that there exist m-Lagrange multipliers $\lambda_1, \ldots, \lambda_m$ such that

$$\operatorname{grad}\Big(f + \sum_{j=1}^{m} \lambda_j g_j\Big) = 0 \quad \text{at } a.$$

Proof. Outline of Proof. Note the constraints

$$g_j(x) = c_j, \qquad j = 1, \ldots, m < n.$$

Using Taylor's expansion for $x = a + \varepsilon h$ yields

$$h^T \nabla g_j(a) = 0$$

for all $j = 1, \ldots, m < n$. Similarly, using $f(a + \varepsilon h) \geq f(a)$ with Taylor's expansion, we obtain

$$h^T \nabla f(a) = 0$$

for all h: $h^T \nabla g_j(a) = 0$ and $j = 1, \ldots, m$. For any $\lambda_1, \ldots, \lambda_m$ in $\mathbb{R}$, we have

$$h^T \big[\nabla f(a) + \sum_{j=1}^{m} \lambda_j \nabla g_j(a)\big] = 0$$

for all h: $h^T \nabla g_j(a) = 0$ and $j = 1, \ldots, m$. This implies that at $x = a$

$$\sum_{i=1}^{m} h_i \Big[\frac{\partial f}{\partial x_i} + \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i}\Big] + \sum_{i=m+1}^{n} h_i \Big[\frac{\partial f}{\partial x_i} + \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i}\Big] = 0 \tag{3.1}$$

for all h: $h^T \nabla g_j(a) = 0$ and $j = 1, \ldots, m$. From linear algebra, we can find $\lambda_1, \ldots, \lambda_m$ such that

$$\frac{\partial f}{\partial x_i} + \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i} = 0 \tag{3.2}$$

for $i = 1, \ldots, m$ at $x = a$. It follows from (3.1)–(3.2) that

$$\sum_{i=m+1}^{n} h_i \Big[\frac{\partial f}{\partial x_i} + \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i}\Big] = 0 \tag{3.3}$$

for all $h = (h_1, \ldots, h_n)$: $h^T \nabla g_j(a) = 0$ and $j = 1, \ldots, m$. For any $(h_{m+1}, \ldots, h_n) \in \mathbb{R}^{n-m}$, there exist $h_1, \ldots, h_m$ such that

$$\sum_{i=1}^{m} h_i \frac{\partial g_j}{\partial x_i} = -\sum_{i=m+1}^{n} h_i \frac{\partial g_j}{\partial x_i} \tag{3.4}$$

for $j = 1, \ldots, m$, which is equivalent to

$$h^T \nabla g_j(a) = 0$$

for $j = 1, \ldots, m$. Since $h_{m+1}, \ldots, h_n$ can choose any number in $\mathbb{R}$ in (3.3), we have

$$\frac{\partial f}{\partial x_i} + \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i} = 0 \tag{3.5}$$

for $i = m + 1, \ldots, n$ at $x = a$.

Combining (3.5) with (3.2), there exist $\lambda_1, \ldots, \lambda_m$ such that

$$\frac{\partial f}{\partial x_i} + \sum_{j=1}^{m} \lambda_j \frac{\partial g_j}{\partial x_i} = 0$$

for $i = 1, \ldots, n$ at $x = a$. This means

$$\operatorname{grad}\left(f + \sum_{j=1}^{m} \lambda_j g_j \right) = 0 \quad \text{at } a.$$

□

Example.

Minimize $f(x) = 1 - x_1^2 - x_2^2$ subject to $g(x) = x_2 - 1 + x_1^2 = 0$.

Using Theorem 1.4.1 with $m = 1$ and $n = 2$, there is a Lagrange multiplier λ such that

$$\operatorname{grad}(f + \lambda g) = 0$$

This is equivalent to

$$-2x_1 + 2\lambda x_1 = 0 \qquad \text{and} \qquad -2x_2 + \lambda = 0.$$

There are three unknowns so we need another equation (i.e. the constraint itself):

$$x_2 - 1 + x_1^2 = 0.$$

Solving these equations we find the following solutions

$$x_1 = 0, \quad x_2 = 1, \quad \lambda = 2$$

and

$$x_1 = \pm\frac{1}{\sqrt{2}}, \quad x_2 = \frac{1}{2}, \quad \lambda = 1.$$

Example.

Find local extrema of

$$f(x) = x_1^3 + x_2^3 + x_3^3$$

where

$$g_1(x) = x_1^2 + x_2^2 + x_3^2 - 5 = 0 \tag{1}$$

$$g_2(x) = x_1^2 + x_2^2 + x_3^2 - 2x_1 - 3 = 0 \tag{2}$$

Lagrangian $L = f + \lambda_1 g_1 + \lambda_2 g_2$:

$$L = x_1^3 + x_2^3 + x_3^3 + \lambda_1(x_1^2 + x_2^2 + x_3^2 - 5) + \lambda_2(x_1^2 + x_2^2 + x_3^2 - 2x_1 - 3)$$

grad $L = 0$ gives

$$3x_1^2 + 2\lambda_1 x_1 + 2\lambda_2(x_1 - 1) = 0 \tag{3}$$

$$3x_2^2 + 2\lambda_1 x_2 + 2\lambda_2 x_2 = 0 \tag{4}$$

$$3x_3^2 + 2\lambda_1 x_3 + 2\lambda_2 x_3 = 0 \tag{5}$$

From (4), $x_2 = 0$ or $3x_2 + 2\lambda_1 + 2\lambda_2 = 0$. From (5), $x_3 = 0$ or $3x_3 + 2\lambda_1 + 2\lambda_2 = 0$.

For $x_3 = 0$, we obtain

$$(1, 2, 0), \quad \lambda_1 = -\tfrac{3}{2}, \quad \lambda_2 = -\tfrac{3}{2}$$

$$(1, -2, 0), \quad \lambda_1 = -\tfrac{3}{2}, \quad \lambda_2 = \tfrac{9}{2}.$$

These and the other solutions ($x_2 = x_3 \neq 0$) above are the critical points of the problem. Are these maxima or minima? We need sufficient conditions to say.

1.5 Distinguish Minima from Maxima

Minimise $f(x_1, x_2, \ldots, x_a)$ subject to

$$\begin{cases} g_1(x_1, \ldots, x_n) = c_1, \\ \vdots \\ g_m(x_1, \ldots, x_n) = c_m \end{cases}$$

Lagrangian

$$L = f + \lambda_1 g_1 + \lambda_2 g_2 + \cdots + \lambda_m g_m.$$

Suppose that $x = a$ is a critical point

$$\# \quad \operatorname{grad} L(a) = 0.$$

Let H_L be the Hessian of L. This means that H_L involves $\lambda_1, \ldots, \lambda_m$ as well as the $a_1, \ldots, a_n$.

$$\# \quad h^T H_L h \geq 0 \quad \text{at } a$$

for all $h \neq 0$ such that $h^T \operatorname{grad} g_i = 0$, for all $1 \leq i \leq m$.

Definition 1.5.1

For $g = (g_1, \ldots, g_m)$, we define $B = \nabla g = \begin{pmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_m}{\partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_n} & \cdots & \frac{\partial g_m}{\partial x_n} \end{pmatrix}$.

Definition 1.5.2

Bordered Hessian $H = \begin{pmatrix} H_L & B \\ B^T & O \end{pmatrix}$ at $x = a$, is a $(m+n) \times (m+n)$ matrix, where O is a $m \times m$ zero matrix.

Definition 1.5.3

Point a at which $\operatorname{grad} L = 0$ and $\det H \neq 0$ is called a nondegenerate critical point.

Definition 1.5.4

h is called tangent vector if $h^T \operatorname{grad} g_i = 0$, for all i.

Theorem 1.5.5

Let a be a non-degenerate critical point for f, subject to $g_i = c_i$, $i = 1, \ldots, m$.

- A necessary and sufficient condition that $x = a$ is a point where f has a local minimum subject to the constraints is that
$$h^T H_L h \geq 0$$
for all tangent vectors h.
- Sufficient condition for a local maximum is that
$$h^T H_L h \leq 0$$
for all tangent vectors h.

Proof. • Step 1.

$$f(a+\varepsilon h) \geq f(a), \quad \forall h : |h| = 1, \ \forall \varepsilon : |\varepsilon| \leq \varepsilon_0$$

with constraint

$$g_j(a+\varepsilon h) = c_j, \quad j = 1, \cdots, m$$

The above problem is equivalent to

$$L(a+\varepsilon h) \geq L(a), \quad \forall h : |h| = 1, \ \forall \varepsilon : |\varepsilon| \leq \varepsilon_0$$

with constraint

$$g_j(a+\varepsilon h) = c_j, \quad j = 1, \cdots, m.$$

Since

$$g_j(a+\varepsilon h) = c_j, \quad j = 1, \cdots, m,$$

we have

$$h^T \nabla g_j(a) = 0.$$

Taylor's expansion yields

$$f(a+\varepsilon h) - f(a) = L(a+\varepsilon h) - L(a)$$

$$= \varepsilon h^T \nabla L(a) + \frac{\varepsilon^2}{2} h^T H_L(a) h + O(\varepsilon^3)$$

for all $h \in \mathbb{R}^n : |h| = 1$ and $h^T \nabla g_j(a) = 0$ with $j = 1, \cdots, m$.

If $h^T H_L(a) h > 0$ for all $h : |h| = 1$ satisfying $h^T \nabla g_j(a) = 0$ with $j = 1, \cdots, m$, there exists $h_0 : |h_0| = 1, \ h_0^T \nabla g_j(a) = 0$ with $j = 1, \cdots, m$ such that

$$h_0^T H_L(a) h_0 = \min h^T H_L(a) h > 0.$$

Then for a sufficiently small ε_0 and $|\varepsilon| \leq \varepsilon_0$,

$$\frac{\varepsilon^2}{2} h^T H_L(a) h + O(\varepsilon^3)$$

$$\geq \frac{\varepsilon^2}{2} h_0^T H_L(a) h_0 + O(\varepsilon^3) > 0.$$

This implies that

$$f(a+\varepsilon h) - f(a) \geq 0$$

with $h^T \nabla g_j(a) = 0$ with $j = 1, \cdots, m$ and $|h| = 1$. This proves that f has a local minimizer at $x = a$. But our assumption that

$$h^T H_L h \geq 0$$

for all tangent vectors h.

• Step 2. Assume there is a $h_0 \neq 0$ with $|h_0| = 1$ and $h_0^T \nabla g_j(a) = 0$ for $j = 1, \cdots, m$ such that

$$h_0^T H_L(a) h_0 = 0.$$

Then we will prove it is impossible. Set $A = H_L(a)$ and

$$B = \nabla g(a) = \left(\frac{\partial g_j}{\partial x_i}\right)_{n \times m}.$$

By the assumption

$$h^T H_L h \geq 0$$

for all tangent vectors $h : h^T \nabla g_j(a) = 0$. This means that h_0 is a minimizer of

$$\min h^T A h = \min \sum_{i,j=1}^{n} h_i A_{ij} h_j$$

with all tangent vectors $h : |h| = 1$ and $h^T \nabla g_j(a) = 0$ for $j = 1, \cdots, m$.

The tangent vector $h : h^T \nabla g_j(a) = 0$ is:

$$h^T B = 0 \quad \text{is equivalent to} \quad B^T h = 0.$$

By Theorem 1.4.1, there exists Lagrangian multiples $\mu_1, \cdots, \mu_m$ such that h_0 satisfies

$$\nabla_h \big(h^T A h + h^T B \mu\big) = 0, \qquad B^T h = 0, \qquad \mu^T = (\mu_1, \cdots, \mu_m)$$

is equivalent to

$$\nabla_h \Big(\sum_{i,j=1}^{n} h_i A_{ij} h_j + h_i \nabla_i g_j \mu_j \Big) = 0, \qquad B^T h = 0.$$

Then it implies

$$2Ah + B\mu = 0, \qquad B^T h = 0$$

which is equivalent to

$$H(a) \begin{pmatrix} 2h \\ \mu \end{pmatrix} = \begin{pmatrix} A & B \\ B^T & O \end{pmatrix} \begin{pmatrix} 2h \\ \mu \end{pmatrix} = \begin{pmatrix} 2Ah + B\mu \\ 2B^T h \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

If

$$\det H(a) = \det \begin{pmatrix} A & B \\ B^T & O \end{pmatrix} \neq 0$$

then $2h = 0$ and $\mu = 0$. This implies that $h_0 = 0$. This proves Step 2. I.e. there exists no $h_0 \neq 0$ with $h_0^T \nabla g_j(a) = 0$ such that $h_0^T H_L(a) h_0 = 0$.

□

To solve constrained problems

1. Construct L
2. Find critical points a where $\operatorname{grad} L = 0$.
3. For $x = a$, check non-degeneracy Bordered Hessian H; i.e. $\det H \neq 0$.
4. Find h in tangent space.
5. Check sign of $h^T H_L h$.

Example.

> Finish above Example: maximise $f : x_1^3 + x_2^3 + x_3^3$ with constraints
>
> $$g_1 : x_1^2 + x_2^2 + x_3^2 - 5 = 0, \quad g_2 : x_1^2 + x_2^2 + x_3^2 - 2x_1 - 3 = 0.$$

We had found the critical points

$$(1, 0, 2), \quad \lambda_1 = -\tfrac{3}{2}, \quad \lambda_2 = -\tfrac{3}{2}$$

$$(1, 0, -2), \quad \lambda_1 = -\tfrac{3}{2}, \quad \lambda_2 = \tfrac{9}{2}$$

$$(1, 2, 0), \quad \lambda_1 = -\tfrac{3}{2}, \quad \lambda_2 = -\tfrac{3}{2}$$

$$(1, -2, 0), \quad \lambda_1 = -\tfrac{3}{2}, \quad \lambda_2 = \tfrac{9}{2}$$

$$L = x_1^3 + x_2^3 + x_3^3 + \lambda_1(x_1^2 + x_2^2 + x_3^2 - 5) + \lambda_2(x_1^2 + x_2^2 + x_3^2 - 2x_1 - 3)$$

$$H_L = \begin{pmatrix} 6x_1 + 2\lambda_1 + 2\lambda_2 & 0 & 0 \\ 0 & 6x_2 + 2\lambda_1 + 2\lambda_2 & 0 \\ 0 & 0 & 6x_3 + 2\lambda_1 + 2\lambda_2 \end{pmatrix}$$

$$\nabla g_1 = \begin{pmatrix} 2x_1 \\ 2x_2 \\ 2x_3 \end{pmatrix}, \qquad \nabla g_2 = \begin{pmatrix} 2x_1 - 2 \\ 2x_2 \\ 2x_3 \end{pmatrix}.$$

Check, for each of the critical points, if

$$H = \begin{pmatrix} H_L & \nabla g_1 & \nabla g_2 \\ \nabla g_1^T & 0 & 0 \\ \nabla g_2^T & 0 & 0 \end{pmatrix}$$

has nonzero determinant.

Check

$$(1,0,2), \quad \lambda_1 = -\tfrac{3}{2}, \quad \lambda_2 = -\tfrac{3}{2}$$

$$H = \begin{pmatrix} 0 & 0 & 0 & 2 & 0 \\ 0 & -6 & 0 & 0 & 0 \\ 0 & 0 & 6 & 4 & 4 \\ 2 & 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \end{pmatrix}$$

$$\det H = -2 \times 6 \times (-4) \times (-2) \times (-16) \neq 0$$

So $(1,0,2)$ is non-degenerate. Find h satisfying $h^T \nabla g_1 = 0$ and $h^T \nabla g_2 = 0$ at critical point:

$$(h_1\ h_2\ h_3) \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} = 0, \qquad (h_1\ h_2\ h_3) \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix} = 0$$

$$\begin{cases} 2h_1 + 4h_3 = 0 \\ 4h_3 = 0 \end{cases} \implies h_1 = h_3 = 0, \quad h_2 = \mu$$

$$h^T = (0, \mu, 0), \qquad \mu \in \mathbb{R}.$$

Check sign of $h^T H_L h$

$$(0\ \mu\ 0) \begin{pmatrix} 0 & 0 & 0 \\ 0 & -6 & 0 \\ 0 & 0 & 6 \end{pmatrix} \begin{pmatrix} 0 \\ \mu \\ 0 \end{pmatrix} = -6\mu^2 < 0.$$

Hence this critical point is local maximum.

Repeat the procedure for other critical points.

Example.

Maximise $xyz = f(x,y,z)$. Subject to

$$g : x + y + z - 1 = 0 \tag{1}$$

$$L(x,y,z) = xyz + \lambda(x + y + z - 1)$$

$$yz + \lambda = 0 \tag{2}$$

$$xz + \lambda = 0 \tag{3}$$

$$xy + \lambda = 0 \tag{4}$$

$$yz = xz = xy = -\lambda$$

Either $x = y$ or $z = 0$ from the first.

If $z = 0$, then $\lambda = 0$ and so $xy = 0$ and at least one of x, y is zero. Note $x = y = z = 0$ does not satisfy (1).

Assume $x = 0$ and $y \neq 0$. Then from (1) $y = 1$ so $(0,1,0)$ is a solution. By symmetry $(1,0,0)$ and $(0,0,1)$ are solutions.

If $z \neq 0$ then $x = y$. Since we have already considered the case $x = 0 = y$ we may assume $x = y \neq 0$. From equations two and three $z = y$ so $x = y = z$.

Substitute on constraint: $3x = 1$

$$\Rightarrow x = y = z = \tfrac{1}{3} \qquad \text{and} \qquad \lambda = -\tfrac{1}{9}.$$

Check sufficiency at the critical point $(1/3, 1/3, 1/3)$ with $\lambda = -1/9$.

$$L = xyz + \lambda(x + y + z - 1)$$

$$H_L = \begin{bmatrix} 0 & z & y \\ z & 0 & x \\ y & x & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & 0 \end{bmatrix}$$

$$\nabla g = \operatorname{grad} g = [1, 1, 1]^T$$

$$H = \begin{bmatrix} H_L & \nabla g \\ \nabla g^T & 0 \end{bmatrix} = \begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} & 1 \\ \frac{1}{3} & 0 & \frac{1}{3} & 1 \\ \frac{1}{3} & \frac{1}{3} & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Noting $C_2 \to C_2 - C_1$, $C_3 \to C_3 - C_1$, where C_r stands for column r, does not change the determinant; we obtain

$$\det H = \begin{vmatrix} 0 & \frac{1}{3} & \frac{1}{3} & 1 \\ \frac{1}{3} & -\frac{1}{3} & 0 & 1 \\ \frac{1}{3} & 0 & -\frac{1}{3} & 1 \\ 1 & 0 & 0 & 0 \end{vmatrix} \neq 0.$$

Hence $(1/3, 1/3, 1/3)$ is a nondegenerate critical point.

Find h on tangent space to g at a

$$h^T \nabla g = 0, \qquad [h_1 \ h_2 \ h_3] \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow h_1 + h_2 + h_3 = 0$$

$$h_1 = \gamma, \quad h_2 = \mu, \quad h_3 = -(\gamma + \mu), \qquad \text{all } \gamma, \mu \in \mathbb{R}$$

Check

$$h^T H_L h = (\gamma \ \mu \ -(\gamma + \mu)) \begin{bmatrix} 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix} \gamma \\ \mu \\ -(\gamma + \mu) \end{bmatrix}$$

$$= (\gamma \ \mu \ -\gamma - \mu) \begin{bmatrix} -\dfrac{\gamma}{3} \\ -\dfrac{\mu}{3} \\ \dfrac{\gamma + \mu}{3} \end{bmatrix}$$

$$= -\frac{\gamma^2}{3} - \frac{\mu^2}{3} - \frac{(\gamma + \mu)^2}{3} < 0$$

Chapter 2

Calculus of Variations

We use (t, x) for the coordinates of a point in the plane and $x = x(t)$ to represent the equation of a plan curve. Let (t_0, x_0) and (t_1, x_1) be two fixed points in the plane.

A curve joining (t_0, x_0) and (t_1, x_1) is represented by $x = x(t)$ for $t_0 \leq t \leq t_1$ with $x(t_0) = x_0$ and $x(t_1) = x_1$.

Theorem 2.0.1

The length of a curve $x = x(t)$ joining (t_0, x_0) and (t_1, x_1) is

$$L[x] = \int_{t_0}^{t_1} \sqrt{1 + \dot{x}^2}\, dt; \qquad \dot{x} = \frac{dx}{dt}.$$

Different type of optimization problems involving integrals of a family of curves :

(1) The classical isoperimetric problem: A curve (closed) of length l. What is the curve which which surrounds the greatest area?

Maximising "area" which depends an curve $x(t)$ (of length l) that encloses it.

(2) Minimizing energy: given a curve $\gamma : x(t)$. What is the curve for which the work (energy) is a minimum for moving P from A to B?

(3) Brachistochrone problem:

A particle is sliding down the curve under gravity. What is the shape of the curve so that the particle falls from A to B in minimum time?

Each of these problems is an optimization problem. However, the "variables" are functions $x(t)$ which define the curve.

2.1 Fixed End-point Problem

Definition 2.1.1

A functional on a curve $x(t)$ joining the point (t_0, x_0) to (t_1, x_1)

$$J[x] = \int_{t_0}^{t_1} f(t, x, \dot{x})\, dt, \quad x(t_0) = x_0, \quad x(t_1) = x_1$$

Here we suppose that f is differentiable with respect to each variable as many times as we require.

Fixed end point problem: Fix (t_0, x_0), (t_1, x_1) and minimise

$$J[x] = \int_{t_0}^{t_1} f(t, x, \dot{x})\, dt,\ x(t_0) = x_0,\ x(t_1) = x_1.$$

That is, find a curve $x^*(t)$ satisfying

$$J[x] \geq J[x^*]$$

for all $x = x(t)$ satisfying the boundary conditions, where x^* is called a **(global) minimizing curve**.

To find a necessary condition, we require some additional assumptions and ideas.

Definition 2.1.2

The class of admissible curves: $x(t)$ twice continuously differentiable, $x(t_0) = x_0,\ x(t_1) = x_1$.

We seek minimising curve $x^*(t)$ in this class.

The problem: Find among the admissible curves, one x^* that gives a local minimum of $J[x]$. That is

$$J[x] \geq J[x^*]$$

for all admissible x close to x^*. What do we mean by "close to" for curves $x(t)$?

Definition 2.1.3: Weak Variation

Let $x^*(t)$ be a curve and $x(t)$ an admissible curve. If $\exists$ small $\varepsilon_1 > 0$ and $\varepsilon_2 > 0$ such that

$$|x^*(t) - x(t)| < \varepsilon_1 \quad |\dot{x}^*(t) - \dot{x}(t)| < \varepsilon_2, \quad \text{for all } t_0 \leq t \leq t_1.$$

then $x(t)$ is said to be a weak variation of $x^*(t)$.

Definition 2.1.4: Weak Local Minimizer

A curve $x^* = x^*(t) : [t_0, t_1] \to \mathbb{R}$ is called "Weakly local minimizer of functional" if there are small $\varepsilon_1 > 0$ and $\varepsilon_2 > 0$ such that

$$J(x^*) \leq J(y)$$

for all curve $y = y(t)$ satisfying the boundary conditions and

$$|x^*(t) - y(t)| < \varepsilon_1 \quad |\dot{x}^*(t) - \dot{y}(t)| < \varepsilon_2, \quad \text{for all } t_0 \leq t \leq t_1.$$

Definition 2.1.5: Strong Variation

$x^*(t), x(t)$ as above. There exists a small $\varepsilon > 0$ such that $|x^*(t) - x(t)| < \varepsilon$ for all $t_0 \leq t \leq t_1$. Then $x(t)$ is called a strong variation of $x^*(t)$.

Definition 2.1.6: Strong local minimizer (minimizing curve)

A curve $x^* = x^*(t) : [t_0, t_1] \to \mathbb{R}$ is called "Strongly local minimizer of functional J" if there is a small ε such that

$$J[x^*] \leq J[y]$$

for all curve $y = y(t)$ satisfying the boundary conditions and

$$|x^*(t) - y(t)| < \varepsilon, \qquad t_0 \leq t \leq t_1.$$

A global minimizing curve is a strong locally minimizing curve. A strong local minimizing curve is a weak local minimizing curve.

A first necessary condition for a "weak local minimum" (i.e. a local minimum with respect to weak variations).

Theorem 2.1.7

In order that $x = x^*(t)$ should be a (weakly) minimizing curve (i.e. minimize $J[x] = \int_{t_0}^{t_1} f(t, x, \dot{x})\, dt,\ x(t_0) = x_0,\ x(t_1) = x_1$) in the class of C^2 functions for the fixed endpoint problem it is necessary that

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\Big(\frac{\partial f}{\partial \dot{x}}\Big) = 0 \tag{4.1}$$

at each point of $x = x^*(t)$.

Maybe examined.

Proof. Outline proof: let $\eta : [t_0, t_1] \to \mathbb{R}$ be a C^1-function with $\eta(t_0) = \eta(t_1) = 0$. Then consider

$$F(\varepsilon) = J(x + \varepsilon\eta) = \int_{t_0}^{t_1} f\big(t, x + \varepsilon\eta, \dot{x} + \varepsilon\dot{\eta}\big)\, dt.$$

By Theorem 1.1.3, $F'(0) = 0$ since $\varepsilon = 0$ is a local minimal point of $F(\varepsilon)$. It follows from the chain rule that

$$\text{(i)} \quad F'(0) = \int_{t_0}^{t_1} \frac{\partial f}{\partial x}\eta + \frac{\partial f}{\partial \dot{x}}\dot{\eta}\, dt = 0.$$

Integration by parts yields

$$(ii) \quad \int_{t_0}^{t_1} \frac{\partial f}{\partial \dot{x}}\eta'\, dt = [f_{\dot{x}}\,\eta]_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{d}{dt}\,(f_{\dot{x}})\,\eta\, dt = -\int_{t_0}^{t_1} \frac{d}{dt}\,(f_{\dot{x}})\,\eta\, dt.$$

due to the fact that $\eta(t_0) = \eta(t_1) = 0$.

From (i)–(ii), we have

$$(iii) \quad \int_{t_0}^{t_1} \Big[\frac{\partial f}{\partial x} - \frac{d}{dt}\Big(\frac{\partial f}{\partial \dot{x}}\Big)\Big]\eta\, dt = 0$$

for all $\eta : [t_0, t_1] \to \mathbb{R}$ of C^1-continuous functions with $\eta(t_0) = \eta(t_1) = 0$.

Choosing $\eta(t) = (t - t_0)^3(t - t_1)^3$ in (iii), we have

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\Big(\frac{\partial f}{\partial \dot{x}}\Big) = 0.$$

□

Definition 2.1.8

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\Big(\frac{\partial f}{\partial \dot{x}}\Big) = 0 \tag{4.1}$$

is called Euler–Lagrange equation. This is an ODE in x, since f is known.

Definition 2.1.9

Solutions of Euler–Lagrange equation are called Extremals.

Definition 2.1.10

$x = x^*(t)$ is a critical point of $J[x] = \int_{t_0}^{t_1} f(t, x, \dot{x})\, dt,\ x(t_0) = x_0,\ x(t_1) = x_1$ if it satisfies

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\Big(\frac{\partial f}{\partial \dot{x}}\Big) = 0.$$

Example.

> Find the extremal of
> $$J[x] = \int_1^2 \dot{x}^2 t^3 \, dt, \qquad x(1) = 0,\ x(2) = 3$$

With $f(t, x, \dot{x}) = \dot{x}^2 t^3$, we have $\dfrac{\partial f}{\partial x} = 0$ and $\dfrac{\partial f}{\partial \dot{x}} = 2\dot{x}t^3$. Euler–Lagrange equation is

$$0 - \frac{d}{dt}\{2\dot{x}t^3\} = 0$$

Thus $\dot{x}t^3 = k$ (a constant).

Separable equation:

$$\int dx = \int \frac{k}{t^3} \, dt + l$$

$$x = \frac{K}{t^2} + l$$

$$x(1) = 0 \Rightarrow K + l = 0$$

$$x(2) = 3 \Rightarrow 3 = \frac{K}{4} - K \Rightarrow K = -4$$

Extremal curve

$$x = 4 - \frac{4}{t^2}.$$

The solution is a curve joining $(1, 0)$ and $(2, 3)$ which is a candidate for minimizing $J[x]$.

Example.

> Brachistochrane: find the path of shortest time on which a particle descents from rest at A to B by gravity.

Let $T = T[x(t)]$ be the time at the position (t, x), where x is vertical and t is horizontal.

Take $A = (0, 0)$ as origin and measure vertically downwards and $B = (a, b)$. The arc-length and the velocity v at the position $(t, x(t))$ are given by

$$s(t) = \int_0^t \sqrt{1 + \dot{x}^2(s)} \, ds, \qquad v = \frac{ds}{dT}.$$

Acceleration is

$$a = \frac{dv}{dT}.$$

Newton's law

$$F = ma,$$

where m is the mass and F is the force. Let g be the gravitation constant. Then

$$m\frac{dv}{dT} = mg\cos\theta = mg\frac{dx}{ds}$$

implying

$$\frac{d^2 s}{dT^2} = g\frac{dx}{ds}.$$

Multiplying $\dfrac{ds}{dT}$ to the above equation yields

$$\frac{ds}{dT}\frac{d^2 s}{dT^2} = g\frac{dx}{ds}\frac{ds}{dT}.$$

Integrating in T, we have

$$\int_0^{T[x(t)]} \frac{ds}{dT}\frac{d^2 s}{dT^2} \, dT = g\int_0^{T[x(t)]} \frac{dx}{ds}\frac{ds}{dT} \, dT.$$

Note that $T(A) = 0$ and $x(0) = 0$. Then

$$\frac{1}{2}\left(\frac{ds}{dT}\right)^2 = gx$$

which implies

$$v = \frac{ds}{dT} = \sqrt{2gx}, \qquad dT = \frac{ds}{\sqrt{2g}}.$$

Therefore, the time $T[x]$ from A to B is

$$T[x] = \int_A^B dT = \int_A^B \frac{dS}{v} = \frac{1}{\sqrt{2g}} \int_0^a \frac{\sqrt{1+\dot{x}^2}}{\sqrt{x}}\, dt,$$

where we used that

$$s(t) = \int_0^t \sqrt{1+\dot{x}^2(\tau)}\, d\tau, \qquad \frac{ds}{dt} = \sqrt{1+\dot{x}^2(t)}.$$

Therefore, we minimize the functional

$$J[x] = \int_0^a \frac{\sqrt{1+\dot{x}^2}}{\sqrt{x}}\, dt.$$

for all curves $x = x(t)$ with $x(0) = 0,\ x(a) = b$.

Integrand does not depend explicitly on t, Then

$$f - \dot{x}\frac{\partial f}{\partial \dot{x}} = \text{const.}$$

For, if $f = f(x, \dot{x})$,

$$\frac{d}{dt}\left(f - \dot{x}\frac{\partial f}{\partial \dot{x}}\right) = \frac{\partial f}{\partial x}\dot{x} + \frac{\partial f}{\partial \dot{x}}\ddot{x} - \dot{x}\frac{\partial f}{\partial x} - \dot{x}\frac{d}{dt}\left(\frac{\partial f}{\partial \dot{x}}\right) = \dot{x}\left[\frac{\partial f}{\partial x} - \frac{d}{dt}\left(\frac{\partial f}{\partial \dot{x}}\right)\right] = 0$$

by Euler–Lagrange.

x^* an extremum of $J[x] = \int_{t_0}^{t_1} f(t, x, \dot{x})\, dt$ with $x = x(t)$, $x(t_0) = x_0$ and $x(t_1) = x_1$

$\iff$

x^* satisfies Euler–Lagrange (E-L): $\frac{\partial f}{\partial x} - \frac{d}{dt}\left(\frac{\partial f}{\partial \dot{x}}\right) = 0$

$\iff$

If f independent of t then E-L, $f - \dot{x}\frac{\partial f}{\partial \dot{x}} = \text{constant}$.

Example.

> Shortest time problem: minimize $\displaystyle\int \frac{\sqrt{1+\dot{x}^2}}{x^{1/2}}\, dt.$

Use the second form of the E – L equation.

$$\left[\frac{1+\dot{x}^2}{x}\right]^{1/2} - \frac{\dot{x}^2}{[x(1+\dot{x}^2)]^{1/2}} = \text{const.} \tag{5.1}$$

$$\{x(1+\dot{x}^2)\}^{-1/2}\{(1+\dot{x}^2) - \dot{x}^2\} = \text{const.}$$

Hence

$$x(1+\dot{x}^2) = c. \tag{5.2}$$

Where $(1+\dot{x}^2)^{1/2}$ occurs, we use a parametric technique to solve the DE. Substitute $\dot{x} = \tan\theta$.

Thus $(1+\dot{x}^2) = (1+\tan^2\theta) = \sec^2\theta = \dfrac{1}{\cos^2\theta}$.

From (5.2),

$$x = c\cos^2\theta. \tag{5.3}$$

This gives x in terms of θ. What we will now do is obtain t in terms of θ.

Differentiate (5.3) and obtain

$$\dot{x} = \frac{d}{d\theta}\big(c\cos^2\theta\big)\frac{d\theta}{dt} = -2c\cos\theta\sin\theta\,\dot{\theta} \quad \text{but} \quad \dot{x} = \tan\theta$$

$$\tan\theta = -2c\cos\theta\sin\theta\,\dot{\theta} \Longleftrightarrow 1 = -2c\dot{\theta}\cos^2\theta = -c\dot{\theta}(1+\cos 2\theta),$$

a first-order separable equation for $\dot{\theta}$, so

$$\int dt = -c\int(1+\cos 2\theta)\,d\theta + K$$

$$t = -c\Big(\theta + \tfrac{1}{2}\sin 2\theta\Big) + K.$$

Finally, setting $k = c/2,\ l = K$ and noting that $x = c\cos^2\theta = c(1+\cos 2\theta)/2$ we obtain

$$x = k(1+\cos 2\theta)$$

$$t = l - k(2\theta + \sin 2\theta).$$

$l,\ k$ are to be found from end conditions.

Important features of this example:

- $f - \dot{x}\frac{\partial f}{\partial \dot{x}} = const. \Longleftrightarrow E\text{–}L\ eqn;$
- $(1+\dot{x}^2)^{1/2}$ *occurs* $\Rightarrow$ *substitute* $\dot{x} = \tan\theta$ *to obtain a parametric form for* $x^*(t)$.

Example.

Find the extremal of

$$J[x] = \int_0^1 \big(tx + \dot{x}^2\big)\,dt, \qquad x(0) = 1, \quad x(1) = 2.75$$

E – L equation:

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\Big[\frac{\partial f}{\partial \dot{x}}\Big] = 0$$

$$0 - \frac{d}{dt}(t + 2\dot{x}) = 0$$

$$\Rightarrow t + 2\dot{x} = 2c \quad \text{(a constant)}$$

$$\Rightarrow \dot{x} = c - \frac{1}{2}t$$

$$\Rightarrow x = ct - \frac{1}{4}t^2 + k$$

$$x(0) = 1 \Rightarrow k = 1; \qquad x(1) = 2.75 = c - 0.25 + 1$$

$$c = 2$$

Extremal

$$x = 2t - \frac{1}{4}t^2 + 1.$$

Example.

$J[x] = \int_0^1 (1+t^2)\dot{x}^2\,dt, \qquad x(0) = \frac{\pi}{2}, \quad x(1) = \pi.$

E – L equation:

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\Big[\frac{\partial f}{\partial \dot{x}}\Big] = 0$$

$$0 - \frac{d}{dt}\{2(1+t^2)\dot{x}\} = 0$$

$$(1+t^2)\dot{x} = c$$

$$\int dx = \int \frac{c}{1+t^2}\,dt + D$$

$$x = c\arctan t + D$$

$$x(0) = \frac{\pi}{2} = D, \qquad x(1) = \pi = c\arctan 1 + \frac{\pi}{2}$$

$$\pi = c\frac{\pi}{4} + \frac{\pi}{2} \quad \Longrightarrow \quad c = 2$$

Extremal

$$x(t) = 2\arctan t + \frac{\pi}{2}.$$

Example.

$J[x] = \int_0^{t_f} x\dot{x}^2\,dt, \quad x(0) = x_0, \quad x(t_f) = x_f$. Here t_f is fixed.

$f = x\dot{x}^2$ does not involve t explicitly. So E–L $\Leftrightarrow f - \dot{x}\frac{\partial f}{\partial \dot{x}} = \text{const.}$

$$\Leftrightarrow x\dot{x}^2 - \dot{x}\cdot 2x\dot{x} = \text{const.} \Leftrightarrow x\dot{x}^2 = c$$

$$\Rightarrow \dot{x}^2 = \frac{c}{x}$$

Let

$$\alpha = \sqrt{c}, \qquad \dot{x} = \frac{\alpha}{\sqrt{x}}$$

$$\int \sqrt{x}\,dx = \int \alpha\,dt + \beta$$

$$\frac{2}{3}x^{3/2} = \alpha t + \beta$$

At $t = 0$:

$$\frac{2}{3}x_0^{3/2} = \beta$$

At $t = t_f$:

$$\frac{2}{3}x_f^{3/2} = \alpha t_f + \frac{2}{3}x_0^{3/2}$$

$$\alpha = \frac{2}{3}\frac{x_f^{3/2} - x_0^{3/2}}{t_f}.$$

Another way to prove Euler–Lagrange equation:

Fixed Endpoint: minimize $J[x] = \displaystyle\int_{t_0}^{t_1} f(t, x, \dot{x})\,dt$ where $(t_0, x(t_0))$ and $(t_1, x(t_1))$ are fixed.

Let $y = x^*(t) + \varepsilon\eta(t)$ be a weak variation starting at $(t_0, x(t_0))$ and ending at $(t_1, x(t_1))$.

We write $O(s)$ for a term T if $|T/s| = |O(s)/s| \le k$ for some constant k and s small. In this case we say the term T is of order big O of s for small s. Thus

$$\lim_{s\to 0}\frac{O(s^2)}{s} = 0.$$

If $x^*(t)$ is a minimizing curve, then we will show that the Euler–Lagrange Equation must be satisfied. Hence x^* satisfies Euler–Lagrange (E–L):

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\Big(\frac{\partial f}{\partial \dot{x}}\Big) = 0.$$

Consider

$$0 \le J[y] - J[x^*] = \Delta J$$

$$= \int_{t_0}^{t_1} f\big(t, x^* + \varepsilon\eta, \dot{x}^* + \varepsilon\dot{\eta}\big)\,dt - \int_{t_0}^{t_1} f\big(t, x^*, \dot{x}^*\big)\,dt$$

$$= \int_{t_0}^{t_1} \Big(f\big(t, x^* + \varepsilon\eta, \dot{x}^* + \varepsilon\dot{\eta}\big) - f\big(t, x^*, \dot{x}^*\big) \Big)\, dt$$

By Taylor's theorem

$$f\big(t, x^* + \varepsilon\eta, \dot{x}^* + \varepsilon\dot{\eta}\big) - f\big(t, x^*, \dot{x}^*\big) = f_x(t, x^*, \dot{x}^*)\varepsilon\eta + f_{\dot{x}}(t, x^*, \dot{x}^*)\varepsilon\dot{\eta} + O(\varepsilon^2).$$

$$\Delta J = \int_{t_0}^{t_1} \big(f_x(t, x^*, \dot{x}^*)\eta + f_{\dot{x}}(t, x^*, \dot{x}^*)\varepsilon\dot{\eta} \big)\, dt + O(\varepsilon^2),$$

Integration by parts

$$\int_{t_0}^{t_1} \dot{u}\, v\, dt = [uv]_{t_0}^{t_1} - \int_{t_0}^{t_1} \dot{v}\, u\, dt.$$

Setting $u = f_{\dot{x}}(t, x^*, \dot{x}^*)$ and $v = \varepsilon\eta$, so that $v = \varepsilon\eta$ and $\dot{u} = \dfrac{d}{dt} f_{\dot{x}}(t, x^*, \dot{x}^*)$, we obtain

$$\int_{t_0}^{t_1} f_{\dot{x}}(t, x^*, \dot{x}^*)\, \varepsilon\dot{\eta}\, dt = [f_{\dot{x}}(t, x^*, \dot{x}^*)\, \varepsilon\eta]_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{d}{dt} f_{\dot{x}}(t, x^*, \dot{x}^*)\, \varepsilon\eta\, dt.$$

Thus,

$$\Delta J = \int_{t_0}^{t_1} \Big(f_x(t, x^*, \dot{x}^*) - \frac{d}{dt} f_{\dot{x}}(t, x^*, \dot{x}^*) \Big)\varepsilon\eta\, dt + \Big[f_{\dot{x}}(t, x^*, \dot{x}^*)\varepsilon\eta \Big]_{t_0}^{t_1} + O(\varepsilon^2)$$

since $\eta(t_0) = 0 = \eta(t_1)$,

$$= \int_{t_0}^{t_1} \Big(f_x(t, x^*, \dot{x}^*) - \frac{d}{dt} f_{\dot{x}}(t, x^*, \dot{x}^*) \Big)\varepsilon\eta\, dt + O(\varepsilon^2)$$

Since this holds for any function η with $\eta(t_0) = 0 = \eta(t_1)$ and any ε, it follows that the Euler–Lagrange equation holds:

$$f_x(t, x^*, \dot{x}^*) - \frac{d}{dt} f_{\dot{x}}(t, x^*, \dot{x}^*) = 0$$

Lemma 2.1.11

If $m(t) : [t_0, t_1] \to \mathbb{R}$ is a continuous function satisfying

$$\int_{t_0}^{t_1} m(t)\eta(t)\, dt = 0$$

for all C^2-functions $\eta(t)$ with $\eta(t_0) = \eta(t_1) = 0$, then $m(t) = 0$ for all $t \in [t_0, t_1]$.

Proof. Assume that there is some $a \in (t_0, t_1)$ so that $m(a) \neq 0\,(> 0)$. By continuity of m, choose a sufficiently small $\varepsilon > 0$ such that $m(t) > 0$ (or $m(t) < 0$) for all $t \in [a - \varepsilon, a + \varepsilon]$. Define

$$\eta(t) = \begin{cases} (t - a + \varepsilon)^3 (a + \varepsilon - t)^3, & t \in [a - \varepsilon, a + \varepsilon], \\ 0, & t \in [t_0, a - \varepsilon] \cup (a + \varepsilon, t_1]. \end{cases}$$

Then

$$\int_{t_0}^{t_1} m(t)\eta(t)\, dt = \int_{a-\varepsilon}^{a+\varepsilon} m(t)\eta(t)\, dt > 0.$$

This is impossible. □

2.2 Problems in Which the End Points Are Not Fixed

Variable endpoint: minimize $J[x] = \int_{t_0}^{t_1} f(t, x, \dot{x})\, dt$ $(t_0, x(t_0))$ is fixed, $(t_1, x(t_1))$ lies on given curve $x = c(t)$. (t_1 **unknown**, and $x(t_1)$ is also unknown)

Let $x^*(t)$ be a minimizing curve and suppose that it intersects the target curve at $t = t_1$. Then x^*

satisfies Euler–Lagrange (E–L):

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\left(\frac{\partial f}{\partial \dot{x}}\right) = 0, \quad \forall t \in [t_0, t_1]$$

Let $y = x^*(t) + \varepsilon\eta(t)$ be a weak variation starting at $(t_0, x(t_0))$ and reaching the curve at $t = t_1 + \triangle\tau$.

Definition 2.2.1

Transversal condition

$$f(t_1) + \big(\dot{c}(t_1) - \dot{x}^*(t_1)\big)\frac{\partial f}{\partial \dot{x}}(t_1) = 0$$

We will show that the following transversality condition must be satisfied.

Proof. Solving $E - L$ eqn, gives two unknown constants. Condition $x(t_0) = x_0$ gives one equation, the transversality condition gives another equation so arbitrary constants can be found.

$$\begin{aligned} y(t_1 + \Delta\tau) &= x^*(t_1 + \Delta\tau) + \epsilon\eta(t_1 + \Delta\tau) \\ &= x^*(t_1) + \Delta\tau\dot{x}^*(t_1) + \epsilon\eta(t_1) + O(\epsilon^2) \\ &= c(t_1 + \Delta\tau) = c(t_1) + \Delta\tau\dot{c}(t_1) + O(\epsilon^2) \end{aligned}$$

Noting $x^*(t_1) = c(t_1)$, ignoring $O(\epsilon^2)$ terms and using the previous equation we obtain

$$\epsilon\eta(t_1) = [\dot{c}(t_1) - \dot{x}^*(t_1)]\Delta\tau. \qquad (1)$$

Consider

$$\begin{aligned} \Delta J &= J[y] - J[x^*] \\ &= \int_{t_0}^{t_1+\Delta\tau} f\big(t, x^* + \epsilon\eta, \dot{x}^* + \epsilon\dot{\eta}\big)\,dt - \int_{t_0}^{t_1} f\big(t, x^*, \dot{x}^*\big)\,dt \\ &= \int_{t_1}^{t_1+\Delta\tau} f\big(t, x^* + \epsilon\eta, \dot{x}^* + \epsilon\dot{\eta}\big)\,dt \\ &+ \int_{t_0}^{t_1} \left(\frac{\partial f}{\partial x}(t, x^*, \dot{x}^*)\eta + \frac{\partial f}{\partial \dot{x}}(t, x^*, \dot{x}^*)\dot{\eta}\right) dt + O(\epsilon^2) \end{aligned}$$

Noting

$$\begin{aligned} &\int_{t_0}^{t_1} \left(\frac{\partial f}{\partial x}(t, x^*, \dot{x}^*)\,\eta + \frac{\partial f}{\partial \dot{x}}(t, x^*, \dot{x}^*)\,\epsilon\dot{\eta}\right) dt \\ &= \int_{t_0}^{t_1} \frac{\partial f}{\partial x}(t, x^*, \dot{x}^*)\,\eta\,dt + \left[\frac{\partial f}{\partial \dot{x}}(t, x^*, \dot{x}^*)\,\epsilon\eta\right]_{t_0}^{t_1} - \int_{t_0}^{t_1} \frac{d}{dt}\frac{\partial f}{\partial \dot{x}}(t, x^*, \dot{x}^*)\,\eta\,dt \\ &= \int_{t_0}^{t_1} \left[\frac{\partial f}{\partial x}(t, x^*, \dot{x}^*) - \frac{d}{dt}\frac{\partial f}{\partial \dot{x}}(t, x^*, \dot{x}^*)\right]\eta\,dt + \left[\frac{\partial f}{\partial \dot{x}}(t, x^*, \dot{x}^*)\,\epsilon\eta\right]_{t_0}^{t_1} \end{aligned}$$

and that

$$\begin{aligned} \Delta J &= \int_{t_1}^{t_1+\Delta\tau} f\big(t, x^* + \epsilon\eta, \dot{x}^* + \epsilon\dot{\eta}\big)dt + \int_{t_0}^{t_1} \left(\frac{\partial f}{\partial x}(t, x^*, \dot{x}^*)\,\eta + \frac{\partial f}{\partial \dot{x}}(t, x^*, \dot{x}^*)\,\epsilon\dot{\eta}\right) dt + O(\epsilon^2) \\ &= f(t_1, x^*, \dot{x}^*)\,\Delta\tau + \int_{t_0}^{t_1} \left[\frac{\partial f}{\partial x}(t, x^*, \dot{x}^*) - \frac{d}{dt}\frac{\partial f}{\partial \dot{x}}(t, x^*, \dot{x}^*)\right]\eta\,dt + \left[\frac{\partial f}{\partial \dot{x}}(t, x^*, \dot{x}^*)\,\epsilon\eta\right]_{t_0}^{t_1} + O(\epsilon^2) \\ &= f(t_1, x^*, \dot{x}^*)\,\Delta\tau + \frac{\partial f}{\partial \dot{x}}(t_1, x^*, \dot{x}^*)\,\epsilon\eta(t_1) + O(\epsilon^2). \end{aligned}$$

Here we used x^* is a solution of $E - L$ eqn and $\eta(t_0) = 0$.

Setting $f(t_1) := f(t_1, x^*(t_1), \dot{x}^*(t_1))$ and substituting for $\epsilon\eta(t_1)$ from (1) we obtain

$$\Delta J = \Delta\tau \left\{ f(t_1) + \left[\dot{c}(t_1) - \dot{x}^*(t_1)\right] \frac{\partial f}{\partial \dot{x}}(t_1) \right\} + O(\epsilon^2).$$

Note $\dfrac{d(\Delta J)}{d\epsilon} = 0$ is called the first variation in J. Noticing

$$\Delta\tau = \frac{\epsilon\,\eta(t_1)}{\dot{c}(t_1) - \dot{x}^*(t_1)}$$

□

Theorem 2.2.2

If $x^*(t)$ is a minimizer, then $x^*(t)$ satisfies transversality and Euler–Lagrange equation.

Definition 2.2.3

An extremal of minimizing $J[x]$ with variable endpoint $\Leftrightarrow$ transversality $+$ $E - L$ equation.

Note that here there were 3 unknowns:

- 2 arbitrary constants from E-L equation
- and T.

Three conditions:

- $x(1) = 0$,
- Transversality condition,
- $x(T) = c(T)$.

Example.

> Find the extremal of
> $$\int_1^T \dot{x}^2 t^3\, dt, \qquad x(1) = 0,$$
> $T > 1$ is finite and $x(T)$ lies on $x = c(t) = \dfrac{2}{t^2} - 3$.

From previous example,

$$x = \frac{k}{t^2} + l \Rightarrow x(1) = 0 \;\Rightarrow\; l = -k \;\Rightarrow\; x = \frac{k}{t^2} - k.$$

Use the transversality condition to find the two unknowns k, T. Now $\dot{x} = -\dfrac{2k}{t^3}$ and $x(t) = c(t) = \dfrac{2}{t^2} - 3$ when $t = T$, so $\dot{c}(t) = -\dfrac{4}{t^3}$.

Transversality condition

$$f(t_1) + \big(\dot{c}(t_1) - \dot{x}^*(t_1)\big) \frac{\partial f}{\partial \dot{x}}(t_1) = 0.$$

With $f(t) = \dot{x}^2 t^3$ and $t_1 = T$:

$$\dot{x}(T)^2 T^3 + 2\dot{x}(T) T^3 \Big[-\frac{4}{T^3} + \frac{2k}{T^3} \Big] = 0,$$

$$-\frac{2k}{T^3} T^3 + 2T^3 \Big[-\frac{4}{T^3} + \frac{2k}{T^3} \Big] = 0,$$

$$2k - 8 = 0 \Rightarrow k = 4.$$

Still have to determine T:

$x(T)$ lies on $x = c(t) = \dfrac{2}{t^2} - 3 \Rightarrow x(T) = \dfrac{2}{T^2} - 3 = \dfrac{4}{T^2} - 4 \Rightarrow \frac{2}{T^2} = 1$

$$\Rightarrow T = \sqrt{2} \quad (T > 1, \text{ so not } -\sqrt{t}).$$

Consequently $x^*(t)$ meets target curve at $T = \sqrt{2}$, when $x(T) = -2$.

2.2.1 Special Forms of Transversality

1. If $x(t_1)$ is fixed and t_1 is variable.

   Target curve here is $x = c(t) = \text{const}$. Thus, $\dot{c}(t) = 0$ and the transversality condition simplifies to

   $$f(t_1) - \dot{x}^*(t_1)\frac{\partial f}{\partial \dot{x}}(t_1) = 0$$

   $$x(t_1) = x_1$$

   Able to solve for 2 constants and t_1

2. t_1 fixed and x_1 variable. This is equivalent to target curve is parallel to x-axis and $\dot{c}(t_1)$ is infinite.

   If $\dot{c}(t_1)$ is infinite, transversality condition:

   $$\frac{1}{\dot{c}(t_1)} f(t_1) + \left[1 - \frac{\dot{x}^*(t_1)}{\dot{c}(t_1)}\right] \frac{\partial f}{\partial \dot{x}}(t_1) = 0.$$

   Let $\dot{c}(t_1) \to \infty$ here, transversality condition simplifies to

   $$\frac{\partial f}{\partial \dot{x}}(t_1) = 0.$$

Example.

Find the extremal of

$$J = \int_0^T \left(x^2 + \dot{x}^2\right) dt$$

when

$$\begin{aligned} &(1)\ x(0) = 1, && T = 2; \\ &(2)\ x(0) = 1, && x(T) = 2; \\ &(3)\ x(0) = 0, && x(T) = 2. \end{aligned}$$

Since f does not involve t explicitly, we could use the other form, but this is easy with the E–L eqn

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\left(\frac{\partial f}{\partial \dot{x}}\right) = 0$$

$$\Rightarrow 2x - \frac{d}{dt}(2\dot{x}) = 0 \Rightarrow \ddot{x} - x = 0.$$

Using the characteristic function,

$$\lambda^2 - 1 = 0 \Rightarrow \lambda = -1, +1$$

Thus,

$$x = Ae^t + Be^{-t}.$$

(1) $x(0) = 1 \;\Rightarrow\; A + B = 1$. Transversality condition is

$$\frac{\partial f}{\partial \dot{x}}(T) = 0 \;\Rightarrow\; 2\dot{x}(T) = 0 \text{ i.e. } \dot{x}(2) = 0$$

$$Ae^2 - Be^{-2} = 0 \;\Rightarrow\; B = Ae^4$$

The extremal is

$$x = \frac{\cosh(t-2)}{\cosh 2} \Rightarrow x(2) = 1/\cosh 2.$$

(2) x(0)=1 ⇒ A+B=1. Transversality condition is

$$f(T) - \dot{x}(T)\frac{\partial f}{\partial \dot{x}}(T) = 0$$

$$\Rightarrow \left(Ae^T + Be^{-T}\right)^2 + \left(Ae^T - Be^{-T}\right)^2 - 2\left(Ae^T - Be^{-T}\right)^2 = 0$$

$$\Rightarrow 2AB + 2AB = 0 \Rightarrow AB = 0.$$

So one of A, B must be zero

$$\Rightarrow x = e^t \quad \text{or} \quad x = e^{-t}.$$

But $x(T) = 2 \Rightarrow 2 = e^T$ or $2 = e^{-T}$. However, if $2 = e^{-T}$, $T < 0$, which is impossible. Hence $2 = e^T$, $T = \ln 2$.

The extremal is $x = e^t$, which has $x = 2$ at time $T = \ln 2$.

Note we use T and t_1 interchangeably so that

$$T = t_1$$

(3) No solution. Because $x := 0$ but $x(T) = 2$.

Note. Variable end at $t = T$ leads to the Transversality condition

$$f(T) + \{\dot{c}(T) - \dot{x}(T)\}\frac{\partial f}{\partial \dot{x}}(T) = 0$$

We consider examples with variable ends which are special cases

Example.

$$J = \int_0^T \left(x^2 + \dot{x}^2\right) dt, \text{ with } x(0) = 0, \quad x(T) = 2.$$

For $f = x^2 + \dot{x}^2$, we have $\ddot{x} - x = 0$. Then $x = Ae^t + Be^{-t}$. By $x(0) = 0$, we have

$$A + B = 0 \quad (1)$$

So $x = c(t) = 2$ at $t = T$ so $\dot{c}(t) = 0$, hence $\dot{c}(T) = 0$. By transversality condition,

$$f(T) - \dot{x}(T)\frac{\partial f}{\partial \dot{x}}(T) = 0 \;\Rightarrow\; AB = 0 \quad (2)$$

But (1) ⇒ $A = -B$ and (2) ⇒ $B^2 = 0$, $B = 0$, so $x = 0$ is the extremal. However, this does not intersect $x = 2$ for any value of T. The problem does not have an extremal.

2.3 Minimizing Curves and Corner Conditions

Let's go back to an earlier example

$$J[x] = \int_1^2 \dot{x}^2 t^3 \, dt, \qquad x(1) = 0, \qquad x(2) = 3.$$

We found that this problem has extremal

$$x = x^*(t) = 4 - \frac{4}{t^2}.$$

Let $y = y(t) = x^*(t) + \eta(t)$ be a variation. Here η is a C^1 curve and $\eta(1) = \eta(2) = 0$, but $\eta(t)$ is not necessarily small. Then the variation on J,

$$\begin{aligned}
\Delta J &= J[y] - J[x^*] \\
&= \int_1^2 t^3\big(8/t^3 + \dot\eta\big)^2 dt - \int_1^2 t^3\big(8/t^3\big)^2 dt \\
&= \int_1^2 \big(16\dot\eta + \dot\eta^2 t^3\big) dt \\
&= [16\eta]_1^2 + \int_1^2 \dot\eta^2 t^3 dt \\
&= 0 + \int_1^2 \dot\eta^2 t^3 dt \geq 0.
\end{aligned}$$

So the extremal $x = x^*(t)$ is a minimizing curve.

However, smooth extremals need not be minimizers. A true minimizer may lie among continuous, piecewise (C^1) curves; this collection is called D_1.

Definition 2.3.1

Piecewise smooth means continuously differentiable except for a possible finite number of corners (jumps in the derivative).

At corners $(t, x^*(t))$ we denote the left hand slope by $\dot x^*(t^-) = p_1$ and the right hand slope by $\dot x^*(t^+) = p_2$.

Example.

Find a minimizing curve for

$$J[x] = \int_0^4 x^2(\dot x - 1)^2\, dt, \qquad x(0) = 0, \quad x(4) = 2.$$

Since the integrand is independent of t,

$$f - \dot x \frac{\partial f}{\partial \dot x} = \text{const.}$$

$$\Rightarrow x^2(\dot x - 1)^2 - \dot x\, 2(\dot x - 1)x^2 = \text{const.}$$

$$\Rightarrow x^2(\dot x^2 - 1) = c, \text{ and } \dot x^2 = \frac{c + x^2}{x^2} \text{ then } \dot x = \frac{\sqrt{c + x^2}}{x}$$

Integrate

$$\int \frac{x\, dx}{\sqrt{c + x^2}} = t + k \Rightarrow \sqrt{c + x^2} = t + k \Rightarrow c + x^2 = (t + k)^2$$

Boundary conditions:

$$\begin{cases} x(0) = 0 \Rightarrow c = k^2 \\ x(4) = 2 \Rightarrow k^2 + 4 = (4 + k)^2 \Rightarrow k = -\frac{3}{2} \end{cases}$$

So $x^2 = t^2 - 3t$ and $(t - \frac{3}{2})^2 - x^2 = \frac{9}{4}$. Something is wrong here: No solution for x when $t < 3$.

However integrand $f = x^2(\dot x - 1)^2 \geq 0 \Rightarrow J \geq 0$ so the absolute minimum $J[x^*] = 0$. A continuous, piecewise C^1 path $x^*(t)$ can make $f := 0$. Therefore $x^*(t)$ must be minimizing. Note that $x^* \in D_1$.

Theorem 2.3.2

In order that a D_1 curve be a minimizing curve for the fixed end point problem, it is necessary that:

1. E–L eqn satisfied between corners, and between corners and endpoints.
2. $\partial f/\partial \dot{x}$ continuous at a corner.
3. $f - \dot{x}\,\partial f/\partial \dot{x}$ continuous at a corner.

The 2nd and 3rd conditions are called corner conditions

Example.

Does the curve

$$(*) \qquad x = \begin{cases} 0, & 0 \le t \le 2 \\ t-2, & 2 \le t \le 4 \end{cases}$$

satisfy these conditions?

1. E–L equation $x^2(\dot{x}^2 - 1) = \text{constant}$. This has the solutions

$$x = \text{const.} \quad \text{or} \quad \dot{x} = 1.$$

   Hence (*) satisfies the E–L equation between each endpoint and corner.

2. Remains to check the corner conditions. Let $p_1 = 0$, $p_2 = 1$ be the slopes just before and just after the corner.

   $\partial f/\partial \dot{x} = 2x^2(\dot{x} - 1)$; before $= 2 \cdot 0^2(p_1 - 1) = 0$, after $= 2 \cdot 0^2(-1 + p_2) = 0$

3. $f - \dot{x}\,\partial f/\partial \dot{x} = -x^2(\dot{x}^2 - 1) = 0$ just before and just after the corner.

Example.

Minimize $\displaystyle\int_0^2 \dot{x}^2(1-\dot{x})^2\,dt,\ x(0) = 0,\ x(2) = 1.$

Euler–Lagrange equation:

$$0 - \frac{d}{dt}\left[2\dot{x}(1-\dot{x})^2 - 2\dot{x}^2(1-\dot{x})\right] = 0.$$

$$\Rightarrow g(\dot{x}) = \text{const where } g(u) = 2u(1-u)^2 - 2u^2(1-u).$$

$$\Rightarrow \dot{x} = \text{const} \ \Rightarrow\ x = kt + l \text{ extremals.}$$

The only smooth extremal satisfying the endpoints is $x^* = t/2$.

To see this is not minimizing: set $y = x^* + \eta(t) = t/2 + \eta(t)$

$$\begin{aligned} J[y] - J[x^*] &= \int_0^2 \left(\tfrac{1}{2} + \dot{\eta}\right)^2 \left(\tfrac{1}{2} - \dot{\eta}\right)^2 dt - \int_0^2 \tfrac{1}{4}\,\tfrac{1}{4}\,dt \\ &= \int_0^2 \left(\left(\tfrac{1}{4} - \dot{\eta}^2\right)^2 - \tfrac{1}{4^2}\right) dt \\ &= \int_0^2 \left(-\tfrac{1}{2}\dot{\eta}^2 + \dot{\eta}^4\right) dt \end{aligned}$$

which is neither > 0 nor < 0 for all $\eta(t)$.

Let us look for a D_1 curve which satisfies the corner conditions and the E–L equation between corners. Since the curve has to satisfy the E–L equation between corners, $\dot{x} = c$ and each segment of the curve is a straight line.

Corner conditions:

$$\partial f/\partial \dot{x} \quad \text{and} \quad f - \dot{x}\,\partial f/\partial \dot{x} \quad \text{continuous at corner}$$

That is, if p_1, p_2 are the slopes on either side of a corner, then

$$g(\dot{x}) := \dot{x}(1-\dot{x})(1-2\dot{x}) \quad \text{and} \quad h(\dot{x}) := \dot{x}^2(1-\dot{x})(3\dot{x}-1)$$

continuous at a corner, means

$$g(p_1) = g(p_2) \quad \text{and} \quad h(p_1) = h(p_2).$$

The only values which satisfy both are

$$p_1 = 0,\ p_2 = 1 \quad \text{or} \quad p_1 = 1,\ p_2 = 0.$$

Fit straight-line segments with those slopes so the broken line meets the endpoints

Do these minimize $J = \displaystyle\int_0^2 \dot{x}^2(1-\dot{x})^2\,dt$? Yes. On both curves, $J = 0$ and these are thus minimizing curves. Note that this is a case where the minimising curve is not unique.

2.4 Isoperimetric Problems

Find a minimizing curve for $J[x]$, while $x = x(t)$ gives another functional $I[x]$ a prescribed value.

Problem: Minimize $J[x] = \int_{t_0}^{t_1} f(t,x,\dot{x})\,dt,\ x(t_0) = x_0,\ x(t_1) = x_1$, subject to the constraint

$$I = \int_{t_0}^{t_1} g(t,x,\dot{x})\,dt = c.$$

The problem is symbolically the same as Lagrange multiplier problems. What is surprising is that the solution is formally the same.

Theorem 2.4.1

For $x = x^*(t)$ to be a solution of the constrained problem, it is necessary that it should be an extremal of

$$J[x] + \lambda I[x] = \int_{t_0}^{t_1} \{f(t,x,\dot{x}) + \lambda g(t,x,\dot{x})\}\,dt$$

for some constant λ.

Proof. Omitted. □

Example.

Minimize

$$J = \int_0^1 \dot{x}^2\,dt, \qquad x(0) = 2, \qquad x(1) = 4,$$

subject to

$$\int_0^1 x\,dt = 1.$$

Lagrangian $J + \lambda I$ is just $\displaystyle\int_0^1 (\dot{x}^2 + \lambda x)\,dt$. Consider the extremals of this functional. Euler–Lagrange equation is

$$\lambda - \frac{d}{dt}(2\dot{x}) = 0$$

i.e. $\ddot{x} = \lambda/2$, or $x = \lambda t^2/4 + kt + l$. From the endpoint conditions:

$$2 = l, \qquad 4 = \lambda/4 + k + 2, \qquad k = 2 - \lambda/4.$$

That is, $x = \lambda t^2/4 + (2 - \lambda/4)t + 2$. Remember to use the constraint $\int_0^1 x\,dt = 1$

$$\lambda/12 + (1 - \lambda/8) + 2 = 1$$

$$\Rightarrow 2 = \lambda/24 \Rightarrow \lambda = 48.$$

Thus, $x = 12t^2 - 10t + 2$.

Solving strategy:

- E–L equation on Lagrangian give differential equation with unknown constants k, l, λ.
- Endpoints gave 2 conditions.
- Constraint gives extra condition.

Example.

> Find, among the set of curves of length π joining $(0,0)$ to $(2,0)$, that which has maximum area between it and the t-axis. That is, maximise $\int_0^2 x\,dt$, subject to $\int_0^2 (1+\dot{x}^2)^{1/2} dt = \pi$ and $x(0) = x(2) = 0$.

Look for extremals of $\int_0^2 \left(x + \lambda(1+\dot{x}^2)^{1/2}\right) dt$. Integrand is independent of t, so the extremals satisfy $f - \dot{x}\,\partial f/\partial \dot{x} = \text{constant}$,

$$x + \lambda(1+\dot{x}^2)^{1/2} - \frac{\dot{x}\ddot{x}\lambda}{(1+\dot{x}^2)^{1.2}} = k.$$

That is,

$$\frac{x(1+\dot{x}^2)^{1/2} + \lambda(1+\dot{x}^2) - \lambda\dot{x}^2}{(1+\dot{x}^2)^{1/2}} = k$$

Put $\dot{x} = \tan\theta$

$$x + \lambda\cos\theta = k$$

Gives x in terms of θ. Differentiate this and equate with $\dot{x} = \tan\theta$, and find θ in terms of t:

$$\dot{x} - \lambda\sin\theta\,\dot{\theta} = 0 \Rightarrow \tan\theta = \lambda\sin\theta\,\dot{\theta} \Rightarrow \lambda\dot{\theta} = \frac{1}{\cos\theta} \quad \text{(separable DE)}$$

Integrate:

$$\int \lambda\cos\theta\,d\theta = t + c \Rightarrow \lambda\sin\theta = t + c$$

Together with $x + \lambda\cos\theta = k$, we get a parametric form of circle.

If θ_0, θ_1 are values of θ at $(0,0)$, $(2,0)$ respectively, the end conditions give

$$k = \lambda\cos\theta_0 = \lambda\cos\theta_1 \quad (1)$$

$$c = \lambda\sin\theta_1 - 2 = \lambda\sin\theta_0 \quad (2)$$

Constraint:

$$\int_0^2 (1+\dot{x}^2)^{1/2} dt = \pi,$$

that is

$$\int_0^2 \sec\theta\,dt = \pi.$$

But "$dt = \lambda\cos\theta\,d\theta$" and so

$$\int_{\theta_0}^{\theta_1} \lambda\,d\theta = \pi \Rightarrow \lambda(\theta_1 - \theta_0) = \pi \quad (3)$$

From (1), $\theta_0 = \pm\theta_1 \quad (\cos\theta_0 = \cos\theta_1)$. If $\theta_1 = \theta_0$, (3) is impossible. Hence $\theta_0 = -\theta_1$ and so

$$\lambda 2\theta_1 = \pi, \quad \theta_1 = \frac{\pi}{2\lambda}.$$

Using (2),

$$\frac{1}{\lambda} = \sin(\pi/2\lambda).$$

This transcendental equation has only 2 solutions, $\lambda = \pm 1$. If $\lambda = 1$, we get a solution in the 4th (t, x)-quadrant. (negative area; a minimum)

So we want the solution given by $\lambda = -1$, for which

$$\theta_1 = -\theta_0 = -\pi/2, \quad k = 0 \quad c = -1.$$

This gives the top half of the circle centred at $(1, 0)$ through $(0, 0)$ and $(2, 0)$.

2.5 Field of extremals

Variational problem:

$$J[y] - J[x^*] = \varepsilon V_1 + \varepsilon^2 V_2 + O(\varepsilon^3),$$

$$V_1 = \int_{t_0}^{t_1} \eta \left(f_x - \frac{d}{dt} f_{\dot{x}} \right) dt, \quad V_2 = \frac{1}{2} \int_{t_0}^{t_1} \left(\eta^2 f_{xx} + 2\eta\dot{\eta} f_{x\dot{x}} + \dot{\eta}^2 f_{\dot{x}\dot{x}} \right) dt.$$

Here $V_1 = 0$ recovers Euler–Lagrange. However, $V_2 > 0$ alone is not sufficient for minimality; extra conditions (e.g., Legendre condition on $f_{\dot{x}\dot{x}}$, absence of conjugate points/Jacobi fields, Weierstrass condition) are needed—this leads to the idea of a field of extremals.

Example.

> $J[x] = \int_0^1 \frac{1}{\dot{x}} dt, \quad x(0) = 0, \quad x(1) = 1.$

Extremal is $x = t = x^*$ and gives $J = 1$. Consider $y = t + \varepsilon\eta$, $\eta(0) = \eta(1) = 0$. Then

$$\Delta J = J[y] - J[t]$$

$$= \int_0^1 \left((1 + \varepsilon\dot{\eta})^{-1} - 1 \right) dt = \int_0^1 \left(-\varepsilon\dot{\eta} + \varepsilon^2\dot{\eta}^2 - \varepsilon^3\dot{\eta}^3 \ldots \right) dt$$

$$= \varepsilon^2 \int_0^1 \dot{\eta}^2 dt + O(\varepsilon^3).$$

Thus, $V_2 > 0$. But $y = \begin{cases} 3t, & 0 \le t \le \frac{1}{2} \\ -t + 2, & \frac{1}{2} \le t \le 1 \end{cases}$ satisfies

$$J[y] = \int_0^{1/2} \frac{dt}{3} + \int_{1/2}^1 (-1) dt = -\frac{1}{3} < J[x^*].$$

However, this is not a D_1 minimizing curve because the corner conditions are not satisfied. So, although $V_2 > 0$, $x = x^* = t$ is not a minimizing curve.

So we have to adopt a more sophisticated approach, using the concept of Hilbert Integral and a Field of Extremals.

Note. Want $\Delta J = J[y] - J[x^*] > 0$ for all $y = y(t)$ satisfying the end conditions.

- ALL $y = x^*(t) + \eta(t)$
- $y(t)$ perhaps in D_1.

The trick is to express ΔJ in another form whose sign is easier to determine. The solution to E–L equation involves two arbitrary constants: two parameter family of curves. Imposing the end conditions gives the constants.

Recall: Example above: $J[x] = \int_0^1 \frac{1}{\dot{x}} dt$ with $x(0) = 0$, $x(1) = 1$.

$$0 - \frac{d}{dt} \left(\frac{-1}{\dot{x}^2} \right) = 0 \Rightarrow \dot{x} = \text{constant} \Rightarrow x = kt + l.$$

Values $k = 1$, $l = 0$ give the extremal through $(0, 0)$ and $(1, 1)$.

Definition 2.5.1: Field of extremals

A 1-parameter family covers the (t, x)-plane simply: through each point passes exactly one curve in the family.

Consider the 1-parameter family of extremals: $x = x_l = t + l$. This family is a simple cover of the plane: through each point passes exactly one curve in the family. It also contain $x = x^* = t$.

Since only one extremal of the family passes through each point, there is a unique value of slope $\dot{x} = p(t, x)$ at each point (t, x). Associated with the field of extremals is a slope function $p(t, x)$ satisfying at each (t, x)

$$\frac{\partial f(t, x, p)}{\partial x} - \frac{d}{dt}\left(\frac{\partial f}{\partial p}(t, x, p)\right) = 0.$$

Example.

> Find a field of extremals for
>
> $$J[x] = \int_1^2 \dot{x}^2 t^3 \, dt$$
>
> with $x(1) = 0$, $x(2) = 3$.

General solution of the E–L equation is

$$x = k/t^2 + l$$

and the extremal satisfying the end conditions is

$$x = 4 - 4/t^2.$$

A 1-parameter family of extremals comparing this is $x = l - 4/t^2$. For $t > 0$ this gives a simple cover of the half plane and so is a field with slope function

$$p(t, x) = 8/t^3.$$

Now $f(t, x, p) = p^2 t^3$, $\partial f/\partial x = 0$ and $\partial f/\partial p = 2pt^3$. Hence

$$\frac{\partial f}{\partial x} - \frac{d}{dx}\left(\frac{\partial f}{\partial p}\right) = 0 - \frac{d}{dt}(2pt^3) = -\frac{d}{dt}(16) = 0.$$

Example.

> Find a field of extremals for
>
> $$J[x] = \int_0^1 \dot{x}^{2m} \, dx, \quad m > 1 \quad \text{an integer}$$
>
> with $x(0) = 1$ and $x(1) = 2$.

The Euler–Lagrange equation is

$$0 - \frac{d}{dt}\left(2m\dot{x}^{2m-1}\right) = 0.$$

This implies that $\dot{x} = \text{constant}$, so

$$x = x^*(t) = kt + l$$

Using $x(0) = 1$ and $x(1) = 2$, we have

$$l = 1 \quad k + l = 2 \Rightarrow k = 1.$$

Let $p = p(t, x) = \dot{x}_l(t) = 1$ be the slope of fields of extremals $x = t + l$. Then

$$f(t, x, p) = p^{2m} = 1, \quad \frac{\partial f}{\partial p} = 2mp^{2m-1} = 2m$$

which satisfies obviously the equation

$$\frac{\partial f}{\partial x} - \frac{d}{dt}\left(\frac{\partial f}{\partial p}\right) = 0$$

for all $(t, x) \in \mathbb{R}$.

2.6 Hilbert's Invariant Integral

Setup: A field of extremals with slope field $p(t, x)$.

- $C^* : x = x^*(t)$ extremal in a field of extremals
- $C : x = x(t)$ any other curve joining the endpoints and covered by the field.

Hilbert's invariant integral:

$$K[x] = \int_{t_0}^{t_1} \left\{ f(t, x, p) + (\dot{x} - p)\frac{\partial f}{\partial p}(t, x, p) \right\} dt.$$

When $C = C^*$, so $\dot{x} = p(t, x^*)$, $K[x^*] = J[x^*]$. In $K[x]$, let

$$u(t, x) = f(t, x, p(t, x)) - p(t, x)\frac{\partial f}{\partial p}(t, x, p(t, x)), \quad v(t, x) = \frac{\partial f}{\partial p}(t, x, p(t, x)).$$

Then

$$\int_{t_0}^{t_1} \dot{x} v \, dt = \int_C v \, dx$$

and we get a line integral:

$$K[x] = \int_C u \, dt + v \, dx.$$

This integral is independent of C but depends on (t_0, x_0), and (t_1, x_1). Thus, $K[x] = K[x^*] = J[x^*]$.

Proof. To see this, we need to show $\partial u/\partial x - \partial v/\partial t = 0$ at each point (t, x). Let x_l be an extremal passing the point (t, x). But, as x, t vary on C

$$\begin{aligned}
\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x}\left[f(t, x, p(t, x)) - p(t, x)\frac{\partial f}{\partial p}(t, x, p(t, x)) \right] \\
&= \frac{\partial f}{\partial x}(t, x, p(t, x)) + \frac{\partial p}{\partial x}(t, x)\frac{\partial f}{\partial p}(t, x, p(t, x)) \\
&\quad - \frac{\partial p}{\partial x}(t, x)\frac{\partial f}{\partial p}(t, x, p(t, x)) - p(t, x)\frac{\partial^2 f}{\partial x \partial p}(t, x, p(t, x)) \\
&\quad - p(t, x)\frac{\partial p}{\partial x}(t, x)\frac{\partial^2 f}{\partial p^2}(t, x, p(t, x)) \\
&= \frac{\partial f}{\partial x}(t, x, p(t, x)) - p(t, x)\frac{\partial^2 f}{\partial x \partial p}(t, x, p(t, x)) \\
&\quad - p(t, x)\frac{\partial p}{\partial x}(t, x)\frac{\partial^2 f}{\partial p^2}(t, x, p(t, x)) \\
&= \frac{\partial f}{\partial x}(t, x_l, p(t, x_l)) - p(t, x_l)\frac{\partial^2 f}{\partial x \partial p}(t, x_l, p(t, x_l)) \\
&\quad - p(t, x_l)\frac{\partial p}{\partial x}(t, x_l)\frac{\partial^2 f}{\partial p^2}(t, x_l, p(t, x_l))
\end{aligned}$$

and

$$\frac{\partial v}{\partial t} = \frac{\partial}{\partial t}\left[\frac{\partial f}{\partial p}(t, x, p(t, x)) \right]$$

$$= \frac{\partial^2 f}{\partial t \partial p}(t, x, p(t, x)) + \frac{\partial^2 f}{\partial p^2}(t, x, p(t, x)) \frac{\partial p}{\partial t}(t, x)$$

$$= \frac{\partial^2 f}{\partial t \partial p}(t, x_l, p(t, x_l)) + \frac{\partial^2 f}{\partial p^2}(t, x_l, p(t, x_l)) \frac{\partial p}{\partial t}(t, x_l),$$

where we use the fact that $(t, x) = (t, x_l(t))$ for an extremal x_l. Using above two identities, we have

$$\frac{\partial u}{\partial x} - \frac{\partial v}{\partial t}(t, x)$$

$$= \frac{\partial f}{\partial x}(t, x_l, p(t, x_l)) - p(t, x_l) \frac{\partial^2 f}{\partial x \partial p}(t, x, p(t, x_l))$$

$$-p(t, x_l) \frac{\partial p}{\partial x}(t, x_l) \frac{\partial^2 f}{\partial p^2}(t, x_l, p(t, x_l))$$

$$-\frac{\partial^2 f}{\partial t \partial p}(t, x_l, p(t, x_l)) - \frac{\partial^2 f}{\partial p^2}(t, x_l, p(t, x_l)) \frac{\partial p}{\partial t}(t, x_l) \tag{6.1}$$

Since $x_l = x_l(t)$ is an extremal in the field of extremals,

$$\frac{dx_l}{dt}(t) = p(t, x_l).$$

Using this equation, we have

$$\frac{d}{dt}\left[\frac{\partial f}{\partial p}(t, x_l, p(t, x_l))\right]$$

$$= \frac{\partial^2 f}{\partial p \partial t}(t, x_l, p(t, x_l)) + \frac{dx_l}{dt} \frac{\partial^2 f}{\partial p \partial x}(t, x_l, p(t, x_l))$$

$$+\frac{d}{dt}\left[p(t, x_l(t))\right] \frac{\partial^2 f}{\partial p^2}(t, x_l, p(t, x_l))$$

$$= \frac{\partial^2 f}{\partial p \partial t}(t, x_l, p(t, x_l)) + \frac{dx_l}{dt} \frac{\partial^2 f}{\partial p \partial x}(t, x_l, p(t, x_l))$$

$$+\left[\frac{\partial p}{\partial t}(t, x_l(t)) + \frac{dx_l}{dt} \frac{\partial p}{\partial x}(t, x_l(t))\right] \frac{\partial^2 f}{\partial p^2}(t, x_l, p(t, x_l))$$

$$= \frac{\partial^2 f}{\partial p \partial t}(t, x_l, p(t, x_l)) + p(t, x_l) \frac{\partial^2 f}{\partial p \partial x}(t, x_l, p(t, x_l))$$

$$+\left[\frac{\partial p}{\partial t}(t, x_l(t)) + p(t, x_l) \frac{\partial p}{\partial x}(t, x_l(t))\right] \frac{\partial^2 f}{\partial p^2}(t, x_l, p(t, x_l)) \tag{6.2}$$

From (6.1) and (6.2), we have

$$\frac{\partial u}{\partial x} - \frac{\partial v}{\partial t}(t, x)$$

$$= \frac{\partial f}{\partial x}(t, x_l, p(t, x_l)) - \frac{d}{dt}\left[\frac{\partial f}{\partial p}(t, x_l, p(t, x_l))\right] = 0$$

since $p(t, x_l) = \dot{x}_l(t)$ is the slope of the field of extremals at the point (t, x).

□

Since $K[x^*] = J[x^*]$, this gives

$$\Delta J = J[x] - J[x^*] = J[x] - K[x^*] = J[x] - K[x].$$

Both integrals are evaluated along C, so:

$$\Delta J = \int_{t_0}^{t_1} \left\{ f(t, x, \dot{x}) - f(t, x, p) - (\dot{x} - p) \frac{\partial f}{\partial p}(t, x, p) \right\} dt$$

where $\dot{x}$ slope of C at (t, x) and p slope of field of extremals at (t, x).

Definition 2.6.1: Weierstrass Excess Function

Let

$$E(t, x, \dot{x}, p) = f(t, x, \dot{x}) - f(t, x, p) - (\dot{x} - p)\frac{\partial f}{\partial p}(t, x, p)$$

denote the integrand in the integral defining ΔJ.

Theorem 2.6.2: Weierstrass Conditions A

In order that the extremal $C^* : x = x^*(t)$ give a strong local minimum to $J[x]$ it is sufficient that

- C^* is a member of a field of extremals;
- $E(t, x, \dot{x}, p) \geq 0 \ \forall (t, x)$ close to C^* and arbitrary values of $\dot{x}$.

Theorem 2.6.3: Weierstrass Conditions B

In order that the extremal $C^* : x = x^*(t)$ give a (global) minimum to $J[x]$ it is sufficient that

- C^* is a member of a field of extremals and the field of extremals cover the whole (t, x)-plan $\mathbb{R}^2$.
- $E(t, x, \dot{x}, p) \geq 0 \ \forall (t, x) \in \mathbb{R}^2$.

Example.

> Example 4. Shortest distance between 2 points is a straight line:
>
> $$\text{minimize} \quad J = \int_0^1 (1 + \dot{x}^2)^{1/2}\, dt, \quad x(0) = 0, \quad x(1) = 1.$$

Extremals are $x = kt + l$, end conditions $k = 1$, $l = 0 \Rightarrow x = t$. First, embed $x = t$ in a field of extremals. Could do $x = t + l$ (with $p(t, x) = 1$) or $x = kt$ ($p = x/t$). Now

$$E(t, x, \dot{x}, p) = f(t, x, \dot{x}) - f(t, x, p) - (\dot{x} - p)\frac{\partial f}{\partial p}(t, x, p)$$

($p = p(t, x)$). Here,

$$E(t, x, \dot{x}, p) = (1 + \dot{x}^2)^{1/2} - (1 + p^2)^{1/2} - p(\dot{x} - p)(1 + p^2)^{-1/2} = (1 + \dot{x}^2)^{1/2} - (1 + p\dot{x})(1 + p^2)^{-1/2}.$$

At a point (t, x) covered by the field, $p = p(t, x)$ has a numerical value. We need to show that $E \geq 0$, for all possible values of $\dot{x}$, at such a point.

- If $1 + p\dot{x} \leq 0$, then $E > 0$.
- $1 + p\dot{x} > 0$; define
$$G = (1 + \dot{x}^2)^{1/2} + (1 + p\dot{x})(1 + p^2)^{-1/2} > 0.$$

$$GE = (1 + \dot{x}^2) - (1 + p\dot{x})^2/(1 + p^2) = \frac{\dot{x}^2 + p^2 - 2p\dot{x}}{1 + p^2} = \frac{(\dot{x} - p)^2}{1 + p^2} \geq 0.$$

So $E \geq 0$ when $1 + p\dot{x} > 0$ and when $1 + p\dot{x} \leq 0$, and by the theorem $x = x^*(t) = t$, is a minimizing curve.

Note. Field of extremals: a family of curves solving the Euler–Lagrange equation that

- contains the endpoint-satisfying extremal $x^*(t)$;
- covers the (t, x)-plane (exactly one curve through each point);
- induces a unique slope function $p(t, x)$ at every (t, x).

Example.

$$\text{Minimizes} \quad J[x] = \int_1^2 \dot{x}^2 t^3 \, dt, \quad x(1) = 0, \quad x(2) = 3.$$

Extremal $x = k/t^2 + l$; satisfying end points, $x = 4 - 4/t^2$.

- Field of extremals $x = l - 4/t^2$; slope of field $p(t, x) = 8/t^3$.
- Excess function

$$f(t, x, \dot{x}) - f(t, x, p) - (\dot{x} - p)\frac{\partial f}{\partial p}(t, x, p)$$

$$E(t, x, \dot{x}, p) = \dot{x}^2 t^3 - p^2 t^3 - (\dot{x} - p)2pt^3 = t^3(\dot{x}^2 - 2p\dot{x} + p^2) = t^3(\dot{x} - p)^2 \geq 0 \quad t \geq 0$$

at each (t, x) for any value of $\dot{x}$. Hence the extremal satisfies the conditions and $x = x^*(t) = 4 - 4/t^2$ is a minimizing curve.

Chapter 3

Optimal Control

3.1 Introduction

Example.

> Level of glucose in bloodstream $x(t)$ is metabolised at a rate $\alpha x(t)$. Change (control) the level of x from $x = a$ at time 0 to $x = c$ at a time T, by infusing at a rate $u(t)$, while minimizing the total amount given to a patient,
> $$\int_0^T u(t)\,dt.$$

That is, minimize $\int_0^T u\,dt$, subject to

$$\dot{x} = -\alpha x + u(t)$$

transfer the system from $x(0) = a$ at time 0 to $x(T) = c$ at time 0, where the time T may be unknown.

Example.

> Imagine a train powered by two rocket engines, one on each side.
>
> - $q(t)$: position at the time t
> - $v(t) = \dot{q}(t)$: velocity at the time t
> - $\alpha(t)$: thrust at the time t from rockets with $-1 \leq \alpha(t) \leq 1$. The sign depends on which engine is firing.

Newton's law from physics: $F = ma$, where F is force, m is mass, and a is acceleration:

$$m\ddot{q}(t) = \alpha(t), \qquad v(t) = \dot{q}(t),$$

with $-1 \leq \alpha(t) \leq 1$. Set

$$\mathbf{x}(t) = \begin{pmatrix} q(t) \\ v(t) \end{pmatrix}.$$

Then

$$\dot{\mathbf{x}}(t) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \mathbf{x}(t) + \frac{\alpha(t)}{m} \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

At the initial time $t = t_0$, $v(t_0) = v_0$ and $q(t_0) = q_0$. At $t = T$, $v(T) = 0$ and $q(T) = 0$. How to fire α such that the train arrive the station $(0, 0)$ in short time; i.e. minimize

$$J = \int_{t_0}^T = (T - t_0).$$

Example.

A space-craft to land on the moon surface

- $h(t)$: height at the time t
- $v(t) = \dot{h}(t)$: velocity at the time t
- $\alpha(t)$: thrust at the time t with $0 \leq \alpha(t) \leq 1$.

Newton's law from physics: $F = ma$, where F is force, m is mass, and a is acceleration; i.e.

$$m\ddot{h} = -gm + \alpha$$

which is equivalent to

$$\dot{v}(t) = -g + \frac{\alpha(t)}{m(t)}, \qquad \dot{h}(t) = v(t).$$

We want to minimize the amount of fuel used up to land on the moon. It is equivalent to maximize the amount of mass remaining when we landed on the surface.
Control Problem: Given the differential equation, control the system from an initial state $\vec{x}(t_0) = \vec{x}_0$ to a given target state $\vec{x}(t_1) = \vec{x}_1$ (t_1 usually free). If this is possible, say the system is **controllable**. If a system is controllable from $\vec{x}_0$ to $\vec{x}_1$ by $\vec{u} \in U$, find the control $\vec{u} = \vec{u}(t)$ minimizing

$$J = \int_{t_0}^{t_1} f_0(\vec{x}, \vec{u})\, dt \quad \text{(cost of transfer)}.$$

Definition 3.1.1

State Equations

$$\dot{x}_i = f_i(\vec{x}, \vec{u}), \quad i = 1, \ldots, 4.$$

Definition 3.1.2

Cost Functional

$$J = \int_{t_0}^{t_1} f_0(\vec{x}, \vec{u})\, dt$$

General Problem: Subject to the DE constraint $\dot{\vec{x}} = \vec{f}(\vec{x}, \vec{u})$, control $x(t)$, an n-dimensional vector, from $\vec{x}_0$ at $t = t_0$ to $\vec{x}_1$ at $t = t_1$ so that the cost functional

$$J = \int_{t_0}^{t_1} f_0(\vec{x}, \vec{u})\, dt$$

is minimized.
Here $\vec{x}_0,\ \vec{x}_1,\ t_0$ are fixed but t_1 is unspecified.

Definition 3.1.3

An admissible control $\vec{u} = \vec{u}^*(t)$ that transfers the system from $\vec{x}_0$ to $\vec{x}_1$ and minimizes J is called an optimal control.

We assume that such a control exists. A first necessary condition analogous to the Euler-Lagrange equation, is needed. The idea is that any control failing this condition could not be optimal. Any control satisfying it *might* be optimal and needs to be looked at more closely.
Given

- State equations $\dot{\vec{x}} = \vec{f}(\vec{x}, \vec{u})$;
- Initial state $\vec{x}(t_0) = \vec{x}_0$;

- Cost functional $J = \int_{t_0}^{t_1} f_0(\vec{x}, \vec{u})\, dt$;
- Set of admissible controls $\vec{u}(t) \in U$ (piecewise continuous.)

Find $\vec{u}^*(t)$ and path $\vec{x}^*(t)$, transferring system from $\vec{x}_0$ to $\vec{x}_1$ such that J is minimized.

3.2 Pontryagin Maximum Principle

Consider a $2-D$ state space

$$\dot{x}_1 = f_1(x_1, x_2, u), \quad \dot{x}_2 = f_2(x_1, x_2, u).$$

Want to go from (x_1^0, x_2^0) at $t = t_0$ to (x_1^1, x_2^1) at time t_1, using piecewise smooth, bounded admissible control functions $u(t)$ so that

$$J = \int_{t_0}^{t_1} f_0(x_1, x_2, u)\, dt \quad \text{minimized.}$$

Definition 3.2.1

Hamiltonian

$$H = \psi_0 f_0(x_1, x_2, u) + \psi_1 f_1(x_1, x_2, u) + \psi_2 f_2(x_1, x_2, u)$$

where $\psi_i = \psi_i(t),\ i = 0, 1, 2$, satisfy $\dot{\psi}_i = -\partial H / \partial x_i, \quad i = 0, 1, 2$ (x_0 defined below).

Note. $\psi_0 = \psi_0(t) = -1$.

Theorem 3.2.2: Pontryagin Maximum Principle

Let $u^*(t)$ be an admissible control with corresponding path $\vec{x}^*(t) = \begin{pmatrix} x_1^* \\ x_2^* \end{pmatrix}$ that takes the system from $(t_0, \vec{x}^0)$ to $(t_1, \vec{x}^1)$ at some (unspecified) time t_1. Then in order that u^* and $\vec{x}^*$ be optimal, it is necessary that $\exists$ a non zero vector $\vec{\psi} = \langle \psi_0, \psi_1, \psi_2 \rangle^T$ such that

$$\dot{\vec{\psi}}_i = -\partial H / \partial x_i, \quad i = 0, 1, 2$$

and a scalar function

$$H(\vec{\psi}, \vec{x}, u) = -f_0 + \psi_1 f_1 + \psi_2 f_2$$

such that

1. $\forall t \in [t_0, t_1],\ H$ attains its maximum as a function of u at $u = u^*(t)$;
2. $H(\vec{\psi}^*, \vec{x}^*, u^*) = 0$ and $\psi_0 \leq 0$ at $t = t_1$, where $\vec{\psi}^*(t)$ is the solution of (PMP) for $u = u^*(t)$.

Furthermore, $H(\vec{\psi}^*(t), \vec{x}^*(t), u^*(t)) = \text{const.}$ and $\psi_0(t) = \text{const.}$, so that $H = 0$ and $\psi_0 \leq 0$ at each point on an optimal trajectory.

Definition 3.2.3

Let x_0 be the solution of the DE

$$\dot{x}_0 = f_0(x_1, x_2, u), \quad x_0(t_0) = 0.$$

That is, $x_0(t)$ measures the cost incurred up to t.

Example.

> Let $\dot{x}_1 = -x_1 + u$. Control the system from $x_1 = a$ at $t = 0$ to $x_1 = b$ at t_1, so that
>
> $$J = \frac{1}{2}\int_0^{t_1} u^2\,dt$$
>
> is minimized.

Either write down the previous equations omitting x_2, f_2, and ψ_2 or let x_2 be a constant by:

$$f_0(x_1, x_2, u) = \frac{1}{2}u^2, \quad f_1(x_1, x_2, u) = -x_1 + u, \quad f_2(x_1, x_2, u) = 0.$$

Hamiltonian

$$H = \psi_0 f_0 + \psi_1 f_1 + \psi_2 f_2 = \psi_0 u^2/2 + \psi_1(-x_1 + u)$$
$$\dot{\psi}_0 = -\partial H/\partial x_0 = 0 \Rightarrow \psi_0 = \text{constant}$$
$$\dot{\psi}_1 = -\partial H/\partial x_1 = \psi_1.$$

Always $\psi_0 =$ constant because H is not a function of x_0 so choose $\psi_0 = -1$ (as long as $\psi_0 < 0$ we can divide everything by ψ_0).
Thus, $\psi_0 = -1$, $\psi_1 = Ae^t$. Now maximize H as a function of u:

$$\partial H/\partial u = \psi_0 u + \psi_1 = -u + \psi_1, \quad \partial^2 H/\partial u^2 = -1.$$

So H is minimized in u by $u = \psi_1$. That is $u = Ae^t$. The corresponding soln for x is

$$\dot{x}_1 = -x_1 + Ae^t \Rightarrow \quad x_1(t) = Be^{-t} + \frac{1}{2}Ae^t.$$

End conditions:

$$x_1(0) = a = B + A/2, \quad x_1(t_1) = b = Be^{-t_1} + Ae^{t_1}/2$$
$$\Rightarrow \quad B = \frac{ae^{t_1} - b}{e^{t_1} - e^{-t_1}}, \quad \frac{A}{2} = \frac{b - ae^{-t_1}}{e^{t_1} - e^{-t_1}}.$$

Free final time $\Rightarrow H \equiv 0$: $H = -AB \Rightarrow AB = 0 \Rightarrow$ either $A = 0$ or $B = 0$.
Assume $a = 2$, $b = 1$. If $B = 0$, $e^{t_1} = 1/2$, impossible for $t_1 > 0$. If $A = 0$, $e^{t_1} = 2$. Then the optimal control is $u = 0$, optimal path $x_1 = 2e^{-t}$, total cost $J = 0$.
So it costs less for the system to go from $x_0 = 2$ to $x_1 = 1$, by exponential decay, than from $x_0 = 1$ to $x_1 = 2$ by forcing. However, if $|u| \le 1$, we could not get from $x_0 = 1$ to $x_1 = 2$ at all. In fact, keeping $u = +1$ just holds the system at the level $x_0 = 1$.

Example.

> $\dot{x}_1 = x_2$, $\dot{x}_2 = u$; control from given initial point (a, b) at $t_0 = 0$ to $\vec{0}$ in as short a time as possible. "Time-optimal control to the origin". Note the system is just $\ddot{x} = u$, $x = a$, $\dot{x} = b$ at $t = t_0 = 0$.

A physical model is a propelled truck $m\ddot{x} = F$, $\quad |F| \le K$, $\quad \ddot{x} = u(t)$, $\quad |u| \le K/m$.

Cost is t_1 and has to be in the form

$$J = \int_0^{t_1} f_0(x_1, x_2, u)\,dt$$

to use the PMP. Choose $f_0 = 1$

$$f_0(x_1, x_2, u) = 1, \quad f_1(x_1, x_2, u) = x_2, \quad f_2(x_1, x_2, u) = u.$$

Hamiltonian

$$H = -1 + \psi_1 x_2 + \psi_2 u$$
$$\begin{cases} \dot{\psi}_1 = -\partial H/\partial x_1 = 0 \\ \dot{\psi}_2 = -\partial H/\partial x_2 = -\psi_1 \end{cases} \Rightarrow \begin{cases} \psi_1 = A \\ \psi_2 = B - At \end{cases}$$

Now maximize H as a function of u, where u is taken as bounded, $|u| \le K/m$. The maximum value of

H if $\psi_2 > 0$ is for $u = K/m$; if $\psi_2 < 0$, max value occurs when $u = -K/m$.

$$u(t) = \frac{K}{m} \text{ sgn } \psi_2(t)$$

That is, the control switches from K/m to $-K/m$ when ψ_2 goes from positive to negative; and switches from $-K/m$ to K/m when ψ_2 changes from negative to positive.
But $\psi_2 = B - At$ can change sign once $\Rightarrow$ only one switch.
Corresponding solution for x_1 and x_2 is as follows: let $u^*(t) = \frac{K}{m}$ sgn $\psi_2(t)$. This is constant between switches, $+K/m$ or $-K/m$:

$$\dot{x}_1 = x_2, \quad \dot{x}_2 = u^* \Rightarrow dx_2/dx_1 = u^*/x_2 \Rightarrow \int x_2 dx_2 = \int u^* dx_1 + \text{ const} \Rightarrow x_2^2 = 2u^* x_1 + \text{ const.}$$

To each of the allowed values of u^* there corresponds a family of parabolas.

3.3 Time Optimal Control of Linear Systems

We consider here systems with two variables $x_1(t), x_2(t)$ describing the state of the system and a single control variable $u(t)$ that is forced to take its values in $|u| \leq 1$. We allow u to be piecewise continuous and let the system be governed by a pair of linear differential equations

$$\dot{x}_1 = ax_1 + bx_2 + lu$$

$$\dot{x}_2 = cx_1 + dx_2 + mu$$

with $|u| \leq 1$ and a, b, c, d, l, m given constants. In matrix notation we have

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{l}\, u \tag{1}$$

where

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad \text{and} \qquad \mathbf{l} = \begin{pmatrix} l \\ m \end{pmatrix}.$$

Problem (The time-optimal problem): Given that system (1) can be controlled from a given initial point $\mathbf{x}(t_0) = \mathbf{x}^0$ to a given target point $\mathbf{x}(t_1) = \mathbf{x}^1$ by an admissible control (piecewise continuous and taking its values from the set $|u| \leq 1$), find the optimal control $u^*(t)$ for which

$$J = \int_{t_0}^{t_1} 1\, dt = t_1 - t_0$$

is minimized.
Solutions will involve PHASE PLANE ANALYSIS.

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0.$$

Revision:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad \mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Solution:

$$\mathbf{x} = \alpha_1 \mathbf{V}_1 e^{\lambda_1 t} + \alpha_2 \mathbf{V}_2 e^{\lambda_2 t},$$

where λ_1, λ_2 are eigenvalues of $\mathbf{A}$ and $\mathbf{V}_1, \mathbf{V}_2$ are the corresponding eigenvectors.

- Case 1. Real nonzero eigenvalues or same sign.
  - $\lambda_2 < \lambda_1 < 0$: stable node.
  - $0 < \lambda_1 < \lambda_2$: unstavle node.
- Case 2. Real, nonzero eigenvalues, of opposite sign $\lambda_1 < 0 < \lambda_2$: saddle (unstable).
- Case 3. imaginary and complex eigenvalues later if time permits

Return to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{l}\,u, \quad f_0 = 1$

1.
$$\begin{aligned} H = \sum \psi_i f_i &= -1 + \psi_1(ax_1 + bx_2 + lu) + \psi_2(cx_1 + dx_2 + mu) \\ &= -1 + \psi_1(ax_1 + bx_2) + \psi_2(cx_1 + dx_2) + u(l\psi_1 + m\psi_2) \end{aligned}$$
$$\begin{cases} \dot{\psi}_1 = -a\psi_1 - c\psi_2 \\ \dot{\psi}_2 = -b\psi_1 - d\psi_2 \end{cases} \quad \Rightarrow \quad \begin{pmatrix} \dot{\psi}_1 \\ \dot{\psi}_2 \end{pmatrix} = \dot{\boldsymbol{\psi}} = -\mathbf{A}^T\boldsymbol{\psi}.$$

2. Maximise H (as a function of u): H is linear in u, so $u^* = \pm 1$, depending on the sign of $l\psi_1 + m\psi_2$:
$$u^* = \operatorname{sgn}(l\psi_1 + m\psi_2).$$
Piecewise continuous controls, switch when $l\psi_1 + m\psi_2$ changes sign (i.e. at zeros of $l\psi_1 + m\psi_2$)
$$S(t) = l\psi_1(t) + m\psi_2(t) \tag{2}$$
That is, they switch from 1 to -1 or from -1 to 1 whenever $S = 0$. For this reason S as defined by (2) is called the switching function.

In the time interval between two zeros of S the control is constant, so the state equations become autonomous:
$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{l}\,u^*, \qquad u^* = +1 \text{ or } -1.$$
If $\det \mathbf{A} \neq 0$, trajectories for $u^* = 1$ have an isolated singularity (equilibrium points) at the solution of
$$ax_1 + bx_2 + l = 0$$
$$cx_1 + dx_2 + m = 0.$$
For $u^* = -1$, the equilibrium is at the solution of
$$ax_1 + bx_2 - l = 0$$
$$cx_1 + dx_2 - m = 0.$$
Behaviour of both families of trajectories is governed by the eigenvalues of the "system matrix" $\mathbf{A}$. The pattern is the same as that of $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$, translated to the equilibrium point.
If $\mathbf{A}$ has real eigenvalues, $-\mathbf{A}^T$ has real eigenvalues also:
$$\mathbf{A}: \ \lambda^2 - (a+d)\lambda + \det \mathbf{A} = 0$$
$$\mathbf{A}^T: \ \lambda^2 + (a+d)\lambda + \det \mathbf{A} = 0.$$
Solution of costate equations has the form
$$\boldsymbol{\psi} = \mathbf{h}\,e^{q_1 t} + \mathbf{K}\,e^{q_2 t}$$
q_1, q_2 eigenvalues of $-\mathbf{A}^T$; $\mathbf{h}, \mathbf{K}$ corresponding eigenvectors.
Hence, switching function is of the form
$$S = l\psi_1 + m\psi_2 = Le^{q_1 t} + Me^{q_2 t}$$
has at most one zero.

Lemma 3.3.1

1. If the eigenvalues of $\mathbf{A}$ are real, then the switching function can have at most one zero.
2. When this is the case the only control sequences that can be optimal are

$$\{1\},\ \{-1\},\ \{1,-1\},\ \{-1,1\}.$$

where for example $\{1\}$ means $u^* = 1$ for $t_0 \le t \le t_1$,

$$\{1,-1\} \text{ means } u^* = \begin{cases} 1, & t_0 \le t \le \tau, \\ -1, & \tau < t \le t_1, \end{cases}$$

and τ is the zero of S.

It can be shown that PMP is both necessary and sufficient for linear time-optimal control problems. Hence, if we can find a control sequence of the type on the lemma, it must be an optimal control.

Example.

$$\dot{x}_1 = -3x_1 + 2x_2 + 5u, \quad \dot{x}_2 = 2x_1 - 3x_2$$

to control from any initial state to the origin in minimum time, $|u| \le 1$. Find the optimal control u^* minimizing

$$J = \int_{t_0}^{t_1} 1\,dt = t_1 - t_0.$$

We have

$$f_0 = 1, \qquad f_1 = -3x_1 + 2x_2 + 5u, \qquad f_2 = 2x_1 - 3x_2.$$

1. $H = -1 + \psi_1(-3x_1 + 2x_2 + 5u) + \psi_2(2x_1 - 3x_2) = -1 + \psi_1(-3x_1 + 2x_2) + \psi_2(2x_1 - 3x_2) + u(5\psi_1)$.

$$\dot{\psi}_1 = -\frac{\partial H}{\partial x_1} = 3\psi_1 - 2\psi_2, \qquad \dot{\psi}_2 = -\frac{\partial H}{\partial x_2} = -2\psi_1 + 3\psi_2$$

$$\dot{\boldsymbol{\psi}} = \begin{pmatrix} 3 & -2 \\ -2 & 3 \end{pmatrix} \boldsymbol{\psi}, \qquad \dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{l}\,u, \quad \mathbf{A} = \begin{pmatrix} -3 & 2 \\ 2 & -3 \end{pmatrix}, \quad \mathbf{l} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}, \qquad \dot{\boldsymbol{\psi}} = -\mathbf{A}^T \boldsymbol{\psi}.$$

2. H is maximized (wrt u) when $u^* = \operatorname{sgn}(5\psi_1) = \pm 1$.
3. Optimal solutions of the state eqns are given by

$$\dot{\mathbf{x}} = \begin{pmatrix} -3 & 2 \\ 2 & -3 \end{pmatrix} \mathbf{x} + \begin{pmatrix} 5 \\ 0 \end{pmatrix} u^*, \qquad u^* = \pm 1.$$

Eigenvalues of

$$\mathbf{A}\ :\ \begin{vmatrix} -3-\lambda & 2 \\ 2 & -3-\lambda \end{vmatrix} = (3+\lambda)^2 - 4 = \lambda^2 + 6\lambda + 5 = 0 \Rightarrow \lambda = -1, -5.$$

The lemma means that there is at most one switch. Since both eigenvalues are negative, the trajectories are a stable node at the equilibrium point.

$$u^* = 1, \text{ equalibrium given by } \quad \left.\begin{matrix} -3x_1 + 2x_2 + 5 = 0 \\ 2x_1 - 3x_2 = 0 \end{matrix}\right\} \Rightarrow \begin{matrix} x_1 = 3 \\ x_2 = 2 \end{matrix}$$

$$u^* = -1, \text{ equalibrium: } \quad \left.\begin{matrix} -3x_1 + 2x_2 - 5 = 0 \\ 2x_1 - 3x_2 = 0 \end{matrix}\right\} \Rightarrow \begin{matrix} x_1 = -3 \\ x_2 = -2 \end{matrix}$$

Eigenvectors of $\mathbf{A}$ are:

$$\begin{pmatrix} -3 & 2 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = -\begin{pmatrix} u \\ v \end{pmatrix}, \quad v = u, \quad \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{V}_1,$$

$$\begin{pmatrix} -3 & 2 \\ 2 & -3 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = -5 \begin{pmatrix} u \\ v \end{pmatrix}, \quad v = -u, \quad \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \mathbf{V}_2.$$